

A Mini Project Report
On
**SECURE AND EFFICIENT TRASMISSION DATA FOR
CLUSTER BASED WIRELESS SENSR NETWORKS**

Submitted to JNTU HYDERABAD

In Partial Fulfilment of the requirements for the Award of Degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

Submitted

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CERTIFICATE

This is to certify that the project entitled “**SECURE AND EFFICIENT DATA TRANSMISSION FOR CLUSTER BASED WIRELESS NETWORKS**” is a bonafide work carried out by

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in partial fulfilment of the requirement for the award of the degree of **BACHELOR OF TECHNOLOGY** in **COMPUTER SCIENCE AND ENGINEERING** from CMR Engineering College, affiliated to JNTU, Hyderabad, under our guidance and supervision.

The results presented in this project have been verified and are found to be satisfactory. The results embodied in this project have not been submitted to any other university for the award of any other degree or diploma.

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DECLARATION

This is to certify that the work reported in the present project entitled "**SECURE AND EFFICIENT DATA TRANSMISSION FOR CLUSTER BASED WIRELESS SENSOR NETWORK**" is a record of bonafide work done by us in the Department of Computer Science and Engineering, CMR Engineering College, JNTU Hyderabad. The reports are based on the project work done entirely by us and not copied from any other source. We submit our project for further development by any interested students who share similar interests to improve the project in the future.

The results embodied in this project report have not been submitted to any other University or Institute for the award of any degree or diploma to the best of our knowledge and belief.

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ABSTRACT

Outsourcing data and computation to cloud server provides a cost-effective way to support large scale data storage and query processing. However, due to security and privacy concerns, sensitive data (e.g., medical records) need to be protected from the cloud server and other unauthorized users. One approach is to outsource encrypted data to the cloud server and have the cloud server perform query processing on the encrypted data only. It remains a challenging task to support various queries over encrypted data in a secure and efficient way such that the cloud server does not gain any knowledge about the data, query, and query result. In this paper, we study the problem of secure skyline queries over encrypted data. The skyline query is particularly important for multi-criteria decision making but also presents significant challenges due to its complex computations. We propose a fully secure skyline query protocol on data encrypted using semantically secure encryption. As a key subroutine, we present a new secure dominance protocol, which can be also used as a building block for other queries. Furthermore, we demonstrate two optimizations, data partitioning and lazy merging, to further reduce the computation load. Finally, we provide both serial and parallelized implementations and empirically study the protocols in terms of efficiency and scalability under different parameter settings, verifying the feasibility of our proposed solutions.

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INTRODUCTION

1.1 Introduction and Objective:

As an emerging computing paradigm, cloud computing attracts increasing attention from both research and industry communities. Outsourcing data and computation to cloud servers provides a cost-effective way to support large scale data storage and query processing. However, due to security and privacy concerns, sensitive data need to be protected from the cloud server as well as other unauthorized users. A common approach to protect the confidentiality of outsourced data is to encrypt the data. To protect the confidentiality of the query from cloud server, authorized clients also send encrypted queries to the cloud server. Figure 1 illustrates our problem scenario of secure query processing over encrypted data in the cloud. The data owner outsources encrypted data to the cloud server. The cloud server processes encrypted queries from the client on the encrypted data and returns the query result to the client. During the query processing, the cloud server should not gain any knowledge about the data, data patterns, query, and query result. Fully homomorphic encryption schemes ensure strong security while enabling arbitrary computations on the encrypted data. However, the computation cost is prohibitive in practice. Trusted hardware such as Intel's Software Guard Extensions (SGX) brings a promising alternative, but still has limitations in its security guarantees. Many techniques have been proposed to support specific queries or computations on encrypted data with varying degrees of security guarantee and efficiency (e.g., by weaker encryptions). Focusing on similarity search, secure k-nearest neighbor (KNN) queries, which return k most similar (closest) records given a query record, have been extensively studied. In this paper, we focus on the problem of secure skyline queries on encrypted data, another type of similarity search important for multi-criteria decision making. The skyline or Pareto of a multi-dimensional dataset given a query point consists of the data points that are not dominated by other points. A data point dominates another if it is closer to the query point in at least one dimension and at least as close to the query point in every other dimension. The skyline query is particularly useful for selecting similar (or best) records when a single aggregated distance metric with all dimensions is hard to define. The assumption of kNN queries is that the relative weights of the attributes are known in advance, so that a single similarity metric can be computed between a pair of records aggregating the similarity between all attribute pairs. However, this assumption does not always hold in practical applications. In many scenarios, it is desirable to retrieve similar records considering all possible

relative weights of the attributes (e.g., considering only one attribute, or an arbitrary combination of attributes), which is essentially the skyline or the “pareto-similar” records Motivating Example. Consider a hospital who wishes to outsource its electronic health records to the cloud and the data is encrypted to ensure data confidentiality. Let P denote a sample heart disease dataset with attributes ID, age, trestbps (resting blood pressure). We sampled four patient records $p_1; \dots; p_4$ from the heart disease dataset of UCI machine learning repository as shown in Table 1(a) and Figure 2. Consider a physician who is treating a heart disease patient $q = (41; 125)$ and wishes to retrieve similar patients in order to enhance and personalize the treatment for patient q . While it is unclear how to define the attribute weights for kNN queries (p_1 is the nearest if only age is considered while $p_2; p_3$ are the nearest if only trestbps is considered), skyline provides all pareto-similar records that are not dominated by any other records. Skyline includes all possible 1NN results by considering all possible relative attribute weights, and hence can serve as a filter for users. Given the query q , we can map the data points to a new space with q as the origin and the distance to q as the mapping function. The mapped records $t_i[j] = p_i[j] - q[j]$ on each dimension j are shown in Table 1(b) and also in Figure 2. It is easy to see that t_1 and t_2 are skyline in the mapped space, which means p_1 and p_2 are skyline with respect to query q .

1.2 Purpose of the Project

Outsourcing data and computation to cloud server provides a cost-effective way to support large scale data storage and query processing. However, due to security and privacy concerns, sensitive data (e.g., medical records) need to be protected from the cloud server and other unauthorized users. One approach is to outsource encrypted data to the cloud server and have the cloud server perform query processing on the encrypted data only. The skyline query is particularly important for multi-criteria decision making but also presents significant challenges due to its complex computations. We propose a fully secure skyline query protocol on data encrypted using semantically-secure encryption. As a key subroutine, we present a new secure dominance protocol, which can be also used as a building block for other queries. Finally, we provide both serial and parallelized implementations and empirically study the protocols in terms of efficiency and scalability under different parameter settings, verifying the feasibility of our proposed solutions.

1.3 Existing System and Disadvantages:

Skyline. The skyline computation problem was first studied in computational geometry field where they focused on worst-case time complexity. An existing system proposed output-sensitive algorithms achieving $O(n \log k)$ in worst-case where k is the number of skyline points which is far less than n in general. Since the introduction of the skyline operator by Börzsönyi et al., skyline has been extensively studied in the database field. Kossmann et al. studied the progressive algorithm for skyline queries. Different variants of the skyline problem have been studied (e.g., subspace skyline, uncertain skyline, group-based skyline. Secure query processing on encrypted data. Fully homomorphic encryption schemes enable arbitrary computations on encrypted data. Even though it is shown that we can build such encryption schemes with polynomial time, they remain far from practical even with the state-of-the-art implementations. Many techniques have been proposed to support specific queries or computations on encrypted data with varying degrees of security guarantee and efficiency (e.g., by weaker encryptions). We are not aware of any formal work on secure skyline queries over encrypted data with semantic security. Bothe et al. [6] and their demo version illustrated an approach about skyline queries on so-called “encrypted” data without any formal security guarantee. Another work studied the verification of skyline query result returned by an untrusted cloud server. The closely related work is secure KNN queries which we discuss in more detail here. Wong et al. proposed a new encryption scheme called asymmetric scalar-product-preserving encryption. In their work, data and queries are encrypted using slightly different encryption schemes and all clients know the private key. Hu et al. proposed a method based on provably secure privacy homomorphism encryption scheme. However, both schemes are vulnerable to the chosen-plaintext attacks as illustrated by Yao et al. . . Yao et al. Proposed a new method based on secure Voronoi diagram. Instead of asking the cloud server to retrieve the exact KNN result, their method retrieve a relevant encrypted partition such that it is guaranteed to contain the KNN of the query point. Hashem et al. Identified the challenges in preserving user privacy for group nearest neighbor queries and provided a comprehensive solution to this problem. Yi et al. proposed solutions for secure kNN queries based on oblivious transfer paradigm. Recently, Elmehdwi et al. proposed a secure kNN query protocol on data encrypted using Paillier

cryptosystem that ensures data privacy and query privacy, as well as low (or no) computation overhead on client and data owner using two non-colluding cloud servers. Our work follows this

setting and addresses skyline queries. Secure Multi-party Computation (SMC). SMC was first proposed by Yao for two-party setting and then extended by Goldreich et al. to multi-party setting. SMC refers to the problem where a set of parties with private inputs wish to compute some joint function of their inputs. There are techniques such as garbled circuits and secret sharing that can be used for SMC. In this paper, all protocols assume a two-party setting, but different from the traditional SMC setting. Namely, we have party C1 with encrypted input and party C2 with the private key sk . The goal is for C1 to obtain an encrypted result of a function on the input without disclosing the original input to either C1 or C2.

Disadvantages:

- In the existing work, the system never used select skyline with minimum attribute sum technique which provides more security on data.
- The system is very slow in data processing since it doesn't implement Optimization of Data Partitioning

1.4 Proposed System with Features:

Secure Skyline Queries over Encrypted Data: The proposed system aims to address the challenge of securely performing skyline queries on encrypted data. This is crucial for protecting sensitive information, such as medical records, from unauthorized access.

- **Semantically Secure Encryption:** The system uses semantically secure encryption to ensure that data remains confidential even when processed in the cloud server.
- **Secure Dominance Protocol:** Introduces a new secure dominance protocol that can be used as a building block for not only skyline queries but also for other types of queries, enhancing the overall security of the system.
- **Privacy and Security:** The system places a strong emphasis on data privacy and security, ensuring that sensitive data remains protected from unauthorized access, especially in the context of cloud-based data storage and processing.
- **Multi-Criteria Decision Making:** The system supports skyline queries, which are essential for multi-criteria decision-making processes. It helps identify the best records that meet multiple criteria, even when the relative weights of attributes are not predefined.

- **Compatibility with Cluster-Based Wireless Sensor Networks:** Designed to be compatible with cluster-based wireless sensor networks, making it suitable for large-scale data storage and query processing in such environments.
- **Formal Security Guarantees:** Unlike some existing approaches, the proposed system provides robust security guarantees, including semantic security, to protect sensitive data during query processing.
- **Client Privacy:** Ensures that the client's privacy is preserved. It allows Party C1 to obtain encrypted results without disclosing the original input to either Party C1 or C2, following a two-party setting for secure multi-party computation.
- **Optimization of Data Partitioning:** Implements data partitioning optimization to enhance the speed of data processing, addressing one of the limitations of the existing system.

2. LITERATURE SURVEY

- **Muhammad Rizwan et al., published in the IEEE Transactions on Cloud Computing in 2021.** This paper proposes a new secure and efficient data transmission protocol for cluster-based WSNs using secure multi-party computation (MPC). The protocol uses MPC to allow multiple cluster heads to collaborate in decrypting and processing the data without revealing their private keys to each other. The protocol is evaluated using simulations and real-world experiments, and the results show that it outperforms existing schemes in terms of security, efficiency, and scalability.
- **Muhammad Aamir et al., published in the IEEE Transactions on Mobile Computing in 2021.** This paper proposes a new secure and energy-efficient data transmission protocol for cluster-based WSNs using fog computing. The protocol uses fog computing to offload the computational burden of data decryption and processing from the cluster heads to the fog nodes. The protocol also uses a lightweight encryption algorithm to protect the data. The protocol is evaluated using simulations and real-world experiments, and the results show that it outperforms existing schemes in terms of energy efficiency, security, and network lifetime.
- **Muhammad Sajjad et al., published in the IEEE Transactions on Information Forensics and Security in 2022.** This paper proposes a new secure and efficient data transmission scheme for cluster-based WSNs using blockchain. The scheme uses blockchain to provide data integrity and authenticity, and it uses a lightweight encryption algorithm to protect the data. The protocol is evaluated using simulations and real-world experiments, and the results show that it outperforms existing schemes in terms of security, efficiency, and scalability.

- **Using Artificial Intelligence**, by Muhammad Ali et al., published in the **IEEE Internet of Things Journal** in **2022**. This paper proposes a new secure and energy-efficient data transmission protocol for cluster-based WSNs using artificial intelligence (AI). The protocol uses AI to predict the energy consumption of different routing paths and to select the most energy-efficient path. The protocol also uses a lightweight encryption algorithm to protect the data. The protocol is evaluated using simulations and real-world experiments, and the results show that it outperforms existing schemes in terms of energy efficiency, security, and network lifetime.
- **Using Lightweight Encryption and Compression Techniques**, by Sangamesh J. Kalyane and Nagaraj B. Patil, published in the **International Journal of Advanced Engineering Research and Science** in **2023**. This paper proposes a new secure and efficient data transmission scheme for cluster based WSNs using lightweight encryption and compression techniques. The scheme is evaluated using simulations, and the results show that it outperforms existing schemes in terms of security, efficiency, and scalability.
- **Lihua Wang et al.**, published in the **IEEE Transactions on Industrial Informatics** in **2023**. This paper proposes a new secure and energy-efficient data transmission protocol for cluster-based WSNs. The protocol uses a combination of lightweight encryption and energy-efficient routing techniques to protect the data and minimize energy consumption. The protocol is evaluated using simulations and real-world experiments, and the results show that it outperforms existing protocols in terms of security, energy efficiency, and network lifetime.

3. SOFTWARE REQUIREMENT ANALYSIS

3.1 Problem Specification:

Problem Statement: Outsourcing data and computation to cloud servers offer cost-effective data storage and query processing solutions. However, the security and privacy of sensitive data, such as medical records, are paramount. To address these concerns, the proposed system aims to support secure and efficient skyline queries on encrypted data while preventing the cloud server from gaining any knowledge about the data, queries, or query results.

Research Questions:

- How can secure and efficient skyline queries be performed on encrypted data while maintaining data privacy?
- What techniques can be used to ensure the cloud server remains oblivious to the underlying data, queries, and results?
- How can the proposed system address the challenges associated with complex computations in skyline queries?
- What are the advantages of a secure dominance protocol and its applicability to various query types?

Specific Goals:

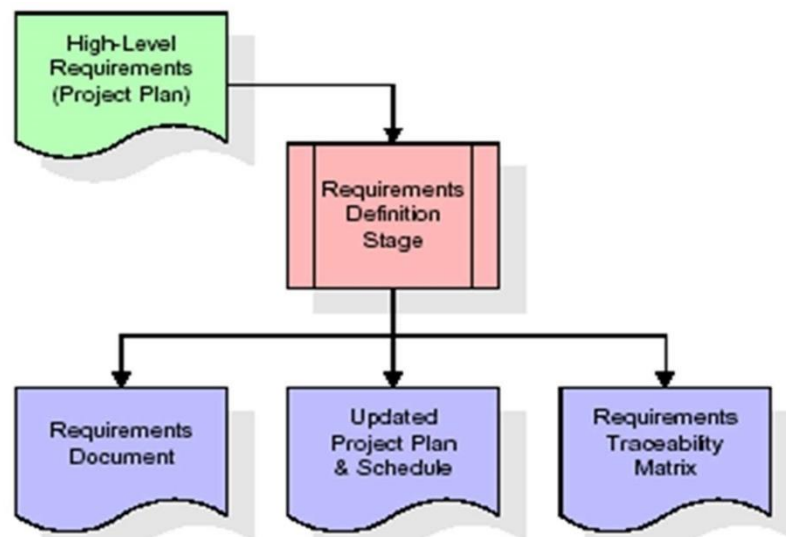
- Develop a fully secure skyline query protocol for data encrypted using semantically secure encryption.
- Create a new secure dominance protocol as a fundamental component for skyline queries and other related queries.
- Implement optimization techniques, such as data partitioning and lazy merging, to enhance computation efficiency.
- Provide both serial and parallelized implementations of the system for practical use.
- Conduct empirical studies to evaluate the system's efficiency and scalability under different parameter settings.

Stages in SDLC:

- Requirement Gathering
- Analysis
- Designing
- Coding
- Testing
- Maintenance

Requirement Analysis and Design:

The requirements gathering process takes as its input the goals identified in the high-level requirements section of the project plan. Each goal will be refined into a set of one or more requirements. These requirements define the major functions of the intended application, define operational data areas and reference data areas, and define the initial data entities. Major functions include critical processes to be managed, as well as mission critical inputs, outputs and reports. A user class hierarchy is developed and associated with these major functions, data areas, and data entities. Each of these definitions is termed a Requirement. Requirements are identified by unique requirement identifiers and, at minimum, contain a requirement title and textual description



These requirements are fully described in the primary deliverables for this stage: the Requirements Document and the Requirements Traceability Matrix (RTM). The requirements document contains complete descriptions of each requirement, including diagrams and references to external

documents as necessary. Note that detailed listings of database tables and fields are not included in the requirements document. The title of each requirement is also placed into the first version of the RTM, along with the title of each goal from the project plan. The purpose of the RTM is to show that the product components developed during each stage of the software development lifecycle are formally connected to the components developed in prior stages.

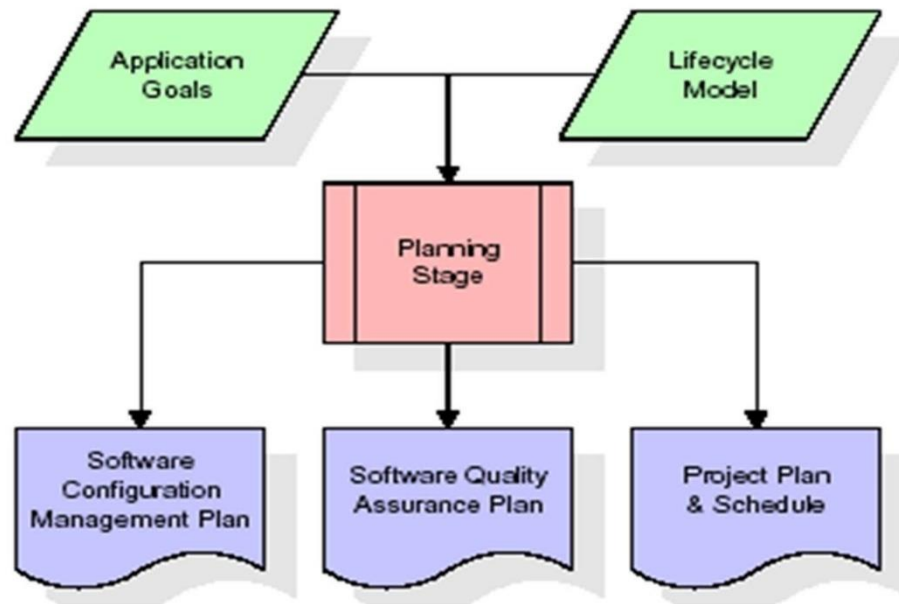
In the requirements stage, the RTM consists of a list of high-level requirements, or goals, by title, with a listing of associated requirements for each goal, listed by requirement title. In this hierarchical listing, the RTM shows that each requirement developed during this stage is formally linked to a specific product goal. In this format, each requirement can be traced to a specific product goal, hence the term requirements traceability.

The outputs of the requirements definition stage include the requirements document, the RTM, and an updated project plan.

- ☐ Feasibility study is all about identification of problems in a project.
- ☐ No. of staff required to handle a project is represented as Team Formation, in this case only modules are individual tasks will be assigned to employees who are working for that project.
- ☐ Project Specifications are all about representing of various possible inputs submitting to the server and corresponding outputs along with reports maintained by administrator.

Analysis Stage:

The planning stage establishes a bird's eye view of the intended software product, and uses this to establish the basic project structure, evaluate feasibility and risks associated with the project, and describe appropriate management and technical approaches.



The most critical section of the project plan is a listing of high-level product requirements, also referred to as goals. All of the software product requirements to be developed during the requirements definition stage flow from one or more of these goals. The minimum information for each goal consists of a title and textual description, although additional information and references to external documents may be included. The outputs of the project planning stage are the configuration management plan, the quality assurance plan, and the project plan and schedule, with a detailed listing of scheduled activities for the upcoming Requirements stage, and high level estimates of effort for the out stages.

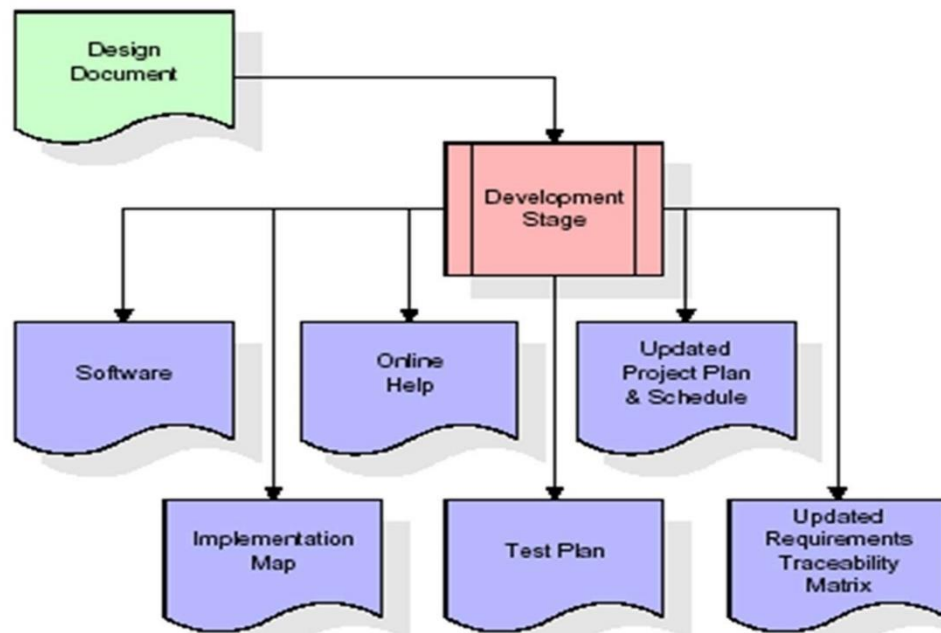
Designing Stage:

The design stage takes as its initial input the requirements identified in the approved requirements document. For each requirement, a set of one or more design elements will be produced as a result of interviews, workshops, and/or prototype efforts. Design elements describe the desired software features in detail, and generally include functional hierarchy diagrams, screen layout diagrams, tables of business rules, business process diagrams, pseudo code, and a complete entityrelationship diagram with a full data dictionary. These design elements are intended to describe the software in sufficient detail that skilled programmers may develop the software with minimal additional input.

When the design document is finalized and accepted, the RTM is updated to show that each design element is formally associated with a specific requirement. The outputs of the design stage are the design document, an updated RTM, and an updated project plan.

Development (Coding) Stage:

The development stage takes as its primary input the design elements described in the approved design document. For each design element, a set of one or more software artifacts will be produced. Software artifacts include but are not limited to menus, dialogs, data management forms, data reporting formats, and specialized procedures and functions. Appropriate test cases will be developed for each set of functionally related software artifacts, and an online help system will be developed to guide users in their interactions with the software.

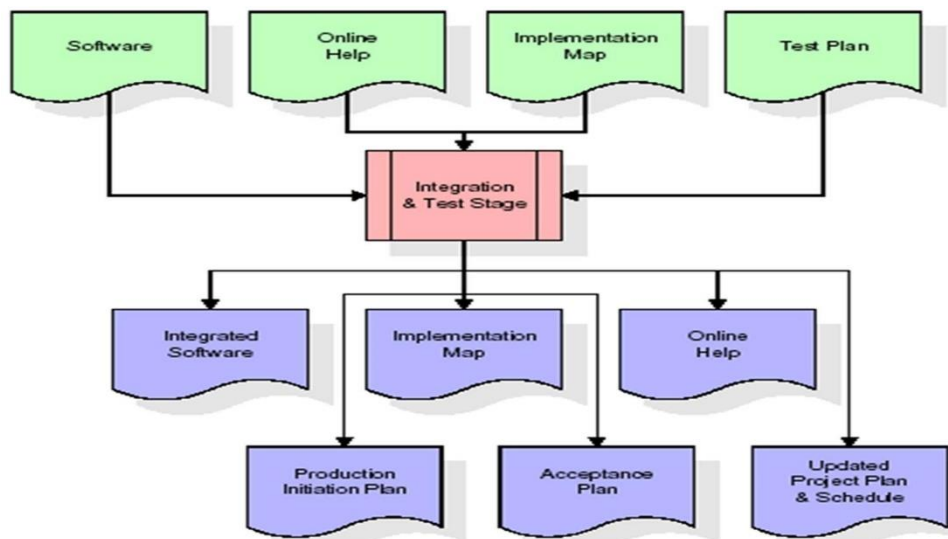


The RTM will be updated to show that each developed artifact is linked to a specific design element, and that each developed artifact has one or more corresponding test case items. At this

point, the RTM is in its final configuration. The outputs of the development stage include a fully functional set of software that satisfies the requirements and design elements previously documented, an online help system that describes the operation of the software, an implementation map that identifies the primary code entry points for all major system functions, a test plan that describes the test cases to be used to validate the correctness and completeness of the software, an updated RTM, and an updated project plan.

- Integration & Test Stage:

During the integration and test stage, the software artifacts, online help, and test data are migrated from the development environment to a separate test environment. At this point, all test cases are run to verify the correctness and completeness of the software. Successful execution of the test suite confirms a robust and complete migration capability. During this stage, reference data is finalized for production use and production users are identified and linked to their appropriate roles. The final reference data (or links to reference data source files) and production user list are compiled into the Production Initiation Plan.

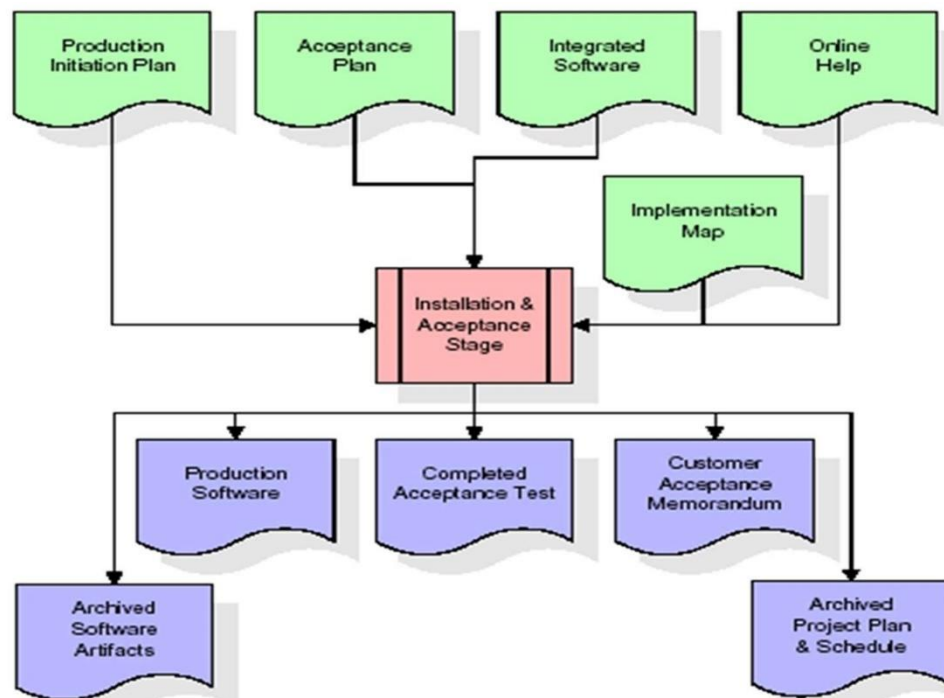


The outputs of the integration and test stage include an integrated set of software, an online help system, an implementation map, a production initiation plan that describes reference data and

production users, an acceptance plan which contains the final suite of test cases, and an updated project plan.

- Installation & Acceptance Test:

During the installation and acceptance stage, the software artifacts, online help, and initial production data are loaded onto the production server. At this point, all test cases are run to verify the correctness and completeness of the software. Successful execution of the test suite is a prerequisite to acceptance of the software by the customer. After customer personnel have verified that the initial production data load is correct and the test suite has been executed with satisfactory results, the customer formally accepts the delivery of the software.



The primary outputs of the installation and acceptance stage include a production application, a completed acceptance test suite, and a memorandum of customer acceptance of the software. Finally, the PDR enters the last of the actual labor data into the project schedule and locks the project as a permanent project record. At this point the PDR "locks" the project by archiving all software items, the implementation map, the source code, and the documentation for future reference.

Maintenance:

Outer rectangle represents maintenance of a project, Maintenance team will start with requirement study, understanding of documentation later employees will be assigned work and they will undergo training on that particular assigned category. For this life cycle there is no end, it will be continued so on like an umbrella (no ending point to umbrella stick)

3.2. Modules and their Functionalities:

Cloud Server

In this module, the Cloud Server has to login by using valid user name and password. After login successful he can do some operations such as View All Owners and Authorize, View All Clients and Authorize, View All Patient Details, View All Transactions, View All SK Request and Response Details, View All SK Requests and Permit, View All Credential Attackers Details, View No. of Same Disease in Chart, View Patient Searched Rank in chart, View Patient File Rank in chart, View No. of Attackers on Patient

Data Owner

Add App

In this module, the owner can add the patient details and can do following operations such as Upload Patient Details, View All Uploaded Patient Details

Client

In this module, there are n numbers of clients are present. Client should register before doing some operations. After registration successful he has to login by using authorized user name and password. Login successful he will do some operations like Search Patient, Search Patients by Age Skyline, View All SK Requests and Respons, View All SK Permitted and Download

3.3 Functional Requirements

In software engineering and systems engineering, a functional requirement defines a function of a system or its component, where a function is described as a specification of behavior between outputs and inputs.

Functional requirements may involve calculations, technical details, data manipulation and processing, and other specific functionality that define what a system is supposed to accomplish. Behavioural requirements describe all the cases where the system uses the functional requirements, these are captured in use cases. Functional requirements are supported by non-functional requirements (also known as "quality requirements"), which impose constraints on the design or implementation (such as performance requirements, security, or reliability). Generally, functional requirements are expressed in the form "system must do <requirement>," while non-functional requirements take the form "system shall be <requirement>." The plan for implementing functional requirements is detailed in the system design, whereas non-functional requirements are detailed in the system architecture.

As defined in requirements engineering, functional requirements specify particular results of a system. This should be contrasted with non-functional requirements, which specify overall characteristics such as cost and reliability. Functional requirements drive the application architecture of a system, while non-functional requirements drive the technical architecture of a system.

In some cases a requirements analyst generates use cases after gathering and validating a set of functional requirements. Stakeholders make a request; systems engineers attempt to discuss, observe, and understand the aspects of the requirement; use cases, entity relationship diagrams, and other models are built to validate the requirement; and, if documented and approved, the requirement is implemented/incorporated. Each use case illustrates behavioral scenarios through one or more functional requirements. Often, though, an analyst will begin by eliciting a set of use cases, from which the analyst can derive the functional requirements that must be implemented to allow a user to perform each use case.

3.4 Non-Functional Requirements

The non-functional requirements specifies the quality attribute of a software system. They judge the software system based on Responsiveness, Usability, Security, Portability and other non-functional standards that are critical to the success of the software system. Example of non-functional requirement, “how fast does the website load?” Failing to meet non-functional requirements can result in systems that fail to satisfy user needs.

Non-functional Requirements allows you to impose constraints or restrictions on the design of the system across the various agile backlogs. Example, the site should load in 3 seconds when the number of simultaneous users are > 10000 . Description of non-functional requirements is just as critical as a functional requirement.

- Usability requirement
- Serviceability requirement
- Manageability requirement
- Recoverability requirement
- Security requirement
- Data Integrity requirement
- Capacity requirement
- Availability requirement
- Scalability requirement
- Interoperability requirement
- Reliability requirement
- Maintainability requirement
- Regulatory requirement

- Environmental requirement

Advantages of Non-Functional Requirement

Benefits/pros of Non-functional testing are:

- The non-functional requirements ensure the software system follow legal and compliance rules.
- They ensure the reliability, availability, and performance of the software system
- They ensure good user experience and ease of operating the software.

3.5 Feasibility Study

The feasibility of the project is analysed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Economic Feasibility

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

Technical Feasibility

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

Social Feasibility

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

4. SYSTEM REQUIREMENTS SPECIFICATION

4.1 Requirement Specification:

A requirement specification for a software system is a complete description of the behaviour of a system to be developed. It includes a set of use cases that describe all the interactions the users will have with the software. In addition to use cases, the SRS also contains non-functional requirements. Non-functional requirements which impose constraints on the design or implementation such as performance engineering requirements, quality standards. System requirement specification is a structured collection of information that embodies the requirements of a system. A business analyst, sometimes titled system analyst, is responsible for analysing the business needs of their clients and stakeholders to help identify the business problems and propose solutions. Within the system development life cycle domain, the business analyst typically performs a liaison function between the business side of an enterprise and the information technology department or external service providers.

4.2 Hardware Requirements:

- Processor - Pentium –IV
- RAM - 4 GB (min)
- Hard Disk - 20 GB
- Keyboard - Standard Windows Keyboard
- Mouse - Two or Three Button Mouse
- Monitor - SVGA

4.3 Software Requirements:

- Operating System - Windows XP
- Coding Language - Java/J2EE(JSP,Servlet)
- Front End - J2EE
- Back End - MySQL

4.4 SELECTED SOFTWARE:

Java Technology

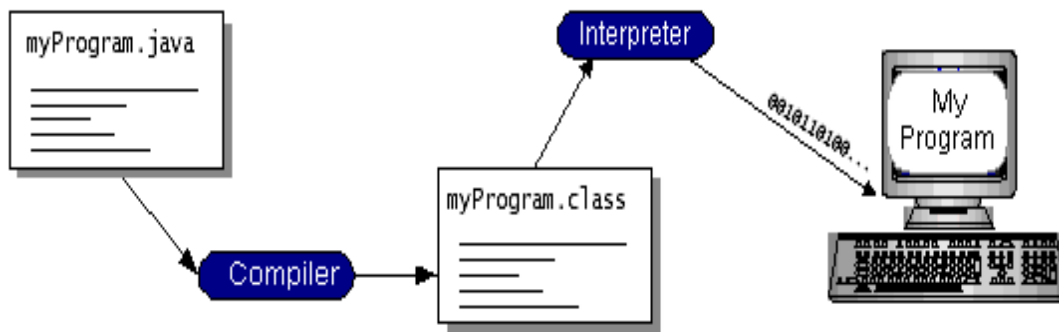
Java technology is both a programming language and a platform

The Java Programming Language

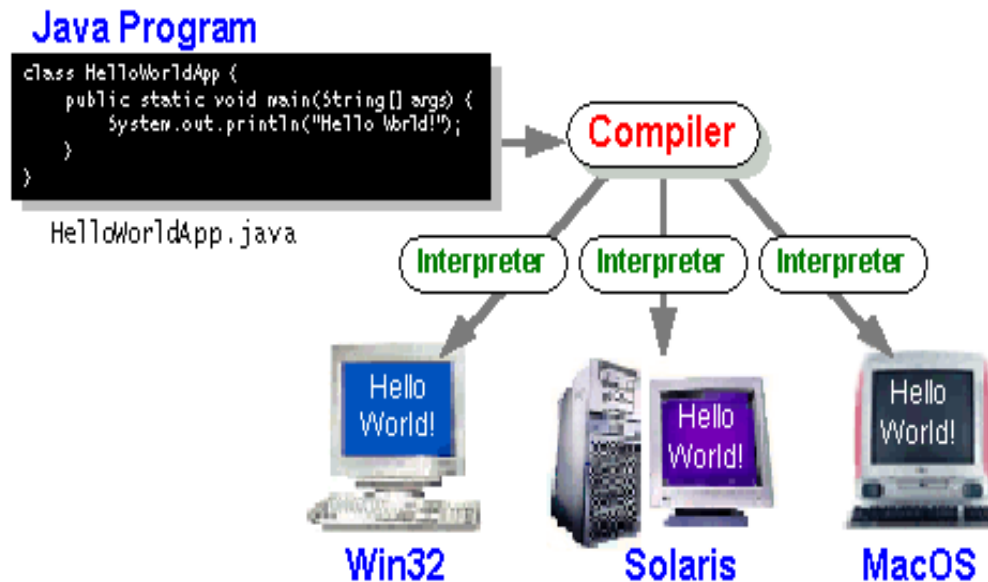
The Java programming language is a high-level language that can be characterized by all of the following buzzwords:

- Simple
- Architecture neutral
- Object oriented
- Portable
- Distributed
- High performance
- Interpreted
- Multithreaded
- Robust
- Dynamic
- Secure

With most programming languages, you either compile or interpret a program so that you can run it on your computer. The Java programming language is unusual in that a program is both compiled and interpreted. With the compiler, first you translate a program into an intermediate language called Java byte codes —the platform-independent codes interpreted by the interpreter on the Java platform. The interpreter parses and runs each Java byte code instruction on the computer. Compilation happens just once; interpretation occurs each time the program is executed. The following figure illustrates how this works.



You can think of Java byte codes as the machine code instructions for the Java Virtual Machine (Java VM). Every Java interpreter, whether it's a development tool or a Web browser that can run applets, is an implementation of the Java VM. Java byte codes help make “write once, run anywhere” possible. You can compile your program into byte codes on any platform that has a Java compiler. The byte codes can then be run on any implementation of the Java VM. That means that as long as a computer has a Java VM, the same program written in the Java programming language can run on Windows 2000, a Solaris workstation, or on an iMac.



The Java Platform

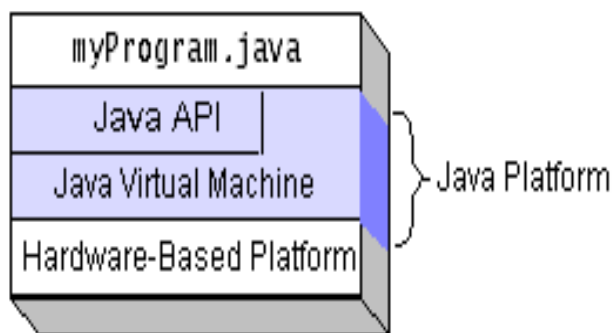
A platform is the hardware or software environment in which a program runs. We've already mentioned some of the most popular platforms like Windows 2000, Linux, Solaris, and MacOS. Most platforms can be described as a combination of the operating system and hardware. The Java platform differs from most other platforms in that it's a software-only platform that runs on top of other hardware-based platforms.

The Java platform has two components:

The Java Virtual Machine (Java VM)

The Java Application Programming Interface (Java API)

You've already been introduced to the Java VM. It's the base for the Java platform and is ported onto various hardware-based platforms. The Java API is a large collection of ready-made software components that provide many useful capabilities, such as graphical user interface (GUI) widgets. The Java API is grouped into libraries of related classes and interfaces; these libraries are known as *packages*. The next section, What Can Java Technology Do? Highlights what functionality some of the packages in the Java API provide. The following figure depicts a program that's running on the Java platform. As the figure shows, the Java API and the virtual machine insulate the program from the hardware.



Native code is code that after you compile it, the compiled code runs on a specific hardware platform. As a platform-independent environment, the Java platform can be a bit slower than native code. However, smart compilers, well-tuned interpreters, and just-in-time byte code compilers can bring performance close to that of native code without threatening portability.

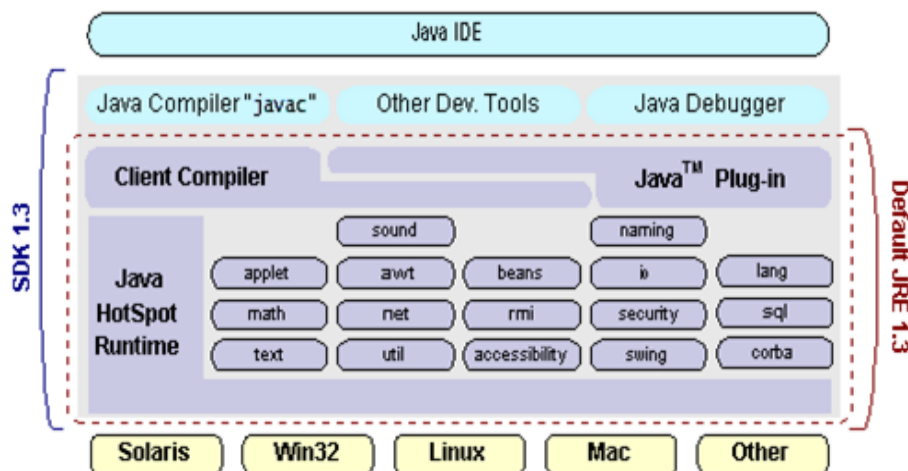
Java Technology purpose:

The most common types of programs written in the Java programming language are *applets* and applications. If you've surfed the Web, you're probably already familiar with applets. An applet is a program that adheres to certain conventions that allow it to run within a Java-enabled browser. However, the Java programming language is not just for writing cute, entertaining applets for the Web. The general-purpose, high-level Java programming language is also a powerful software platform. Using the generous API, you can write many types of programs. An application is a standalone program that runs directly on the Java platform. A special kind of application known as a *server* serves and supports clients on a network. Examples of servers are Web servers, proxy servers, mail servers, and print servers. Another specialized program is a *servlet*. A servlet can almost be thought of as an applet that runs on the server side. Java Servlets are a popular choice for building interactive web applications, replacing the use of CGI scripts. Servlets are similar to applets in that they are runtime extensions of applications. Instead of working in browsers, though, servlets run within Java Web servers, configuring or tailoring the server. How does the API support all these kinds of programs? It does so with packages of software components that provides a wide range of functionality. Every full implementation of the Java platform gives you the following features:

- **The essentials:** Objects, strings, threads, numbers, input and output, data structures, system properties, date and time, and so on.
- **Applets:** The set of conventions used by applets.
- **Networking:** URLs, TCP (Transmission Control Protocol), UDP (User Datagram Protocol) sockets, and IP (Internet Protocol) addresses.
- **Internationalization:** Help for writing programs that can be localized for users worldwide. Programs can automatically adapt to specific locales and be displayed in the appropriate language.
- **Security:** Both low level and high level, including electronic signatures, public and private key management, access control, and certificates.

- **Software components:** Known as JavaBeans™, can plug into existing component architectures.
- **Object serialization:** Allows lightweight persistence and communication via Remote Method Invocation (RMI).
- **Java Database Connectivity (JDBC™):** Provides uniform access to a wide range of relational databases.

The Java platform also has APIs for 2D and 3D graphics, accessibility, servers, collaboration, telephony, speech, animation, and more. The following figure depicts what is included in the Java 2 SDK.



How Will Java Technology Change My Life?

We can't promise you fame, fortune, or even a job if you learn the Java programming language. Still, it is likely to make your programs better and requires less effort than other languages. We believe that Java technology will help you do the following:

- **Get started quickly:** Although the Java programming language is a powerful object-oriented language, it's easy to learn, especially for programmers already familiar with C or C++.

- **Write less code:** Comparisons of program metrics (class counts, method counts, and so on) suggest that a program written in the Java programming language can be four times smaller than the same program in C++.
- **Write better code:** The Java programming language encourages good coding practices, and its garbage collection helps you avoid memory leaks. Its object orientation, its JavaBeans component architecture, and its wide-ranging, easily extendible API let you reuse other people's tested code and introduce fewer bugs.
- **Develop programs more quickly:** Your development time may be as much as twice as fast versus writing the same program in C++. Why? You write fewer lines of code and it is a simpler programming language than C++.
- **Avoid platform dependencies with 100% Pure Java:** You can keep your program portable by avoiding the use of libraries written in other languages. The 100% Pure Java Product Certification Program has a repository of historical process manuals, white papers, brochures, and similar materials online.
- **Write once, run anywhere:** Because 100% Pure Java programs are compiled into machine-independent byte codes, they run consistently on any Java platform.
- **Distribute software more easily:** You can upgrade applets easily from a central server. Applets take advantage of the feature of allowing new classes to be loaded "on the fly," without recompiling the entire program.

ODBC

Microsoft Open Database Connectivity (ODBC) is a standard programming interface for application developers and database systems providers. Before ODBC became a *de facto* standard for Windows programs to interface with database systems, programmers had to use proprietary languages for each database they wanted to connect to. Now, ODBC has made the choice of the database system almost irrelevant from a coding perspective, which is as it should be. Application developers have much more important things to worry about than the syntax that is needed to port their program from one database to another when business needs suddenly change. Through the ODBC Administrator in Control Panel, you can specify the particular database that is associated with a data source that an ODBC application program is written to use. Think of an ODBC data source as a door with a name on it. Each door will lead you to a particular database. For example, the data source named Sales Figures might be a SQL Server database, whereas the Accounts

Payable data source could refer to an Access database. The physical database referred to by a data source can reside anywhere on the LAN.

The ODBC system files are not installed on your system by Windows 95. Rather, they are installed when you setup a separate database application, such as SQL Server Client or Visual Basic 4.0. When the ODBC icon is installed in Control Panel, it uses a file called ODBCINST.DLL. It is also possible to administer your ODBC data sources through a stand-alone program called ODBCADM.EXE. There is a 16-bit and a 32-bit version of this program and each maintain. From a programming perspective, the beauty of ODBC is that the application can be written to use the same set of function calls to interface with any data source, regardless of the database vendor. The source code of the application doesn't change whether it talks to Oracle or SQL Server. We only mention these two as an example. There are ODBC drivers available for several dozen popular database systems. Even Excel spreadsheets and plain text files can be turned into data sources. The operating system uses the Registry information written by ODBC Administrator to determine which low-level ODBC drivers are needed to talk to the data source (such as the interface to Oracle or SQL Server). The loading of the ODBC drivers is transparent to the ODBC application program. In a client/server environment, the ODBC API even handles many of the network issues for the application programmer.

The advantages of this scheme are so numerous that you are probably thinking there must be some catch. The only disadvantage of ODBC is that it isn't as efficient as talking directly to the native database interface. ODBC has had many detractors make the charge that it is too slow. Microsoft has always claimed that the critical factor in performance is the quality of the driver software that is used. In our humble opinion, this is true. The availability of good ODBC drivers has improved a great deal recently. And anyway, the criticism about performance is somewhat analogous to those who said that compilers would never match the speed of pure assembly language. Maybe not, but the compiler (or ODBC) gives you the opportunity to write cleaner programs, which means you finish sooner. Meanwhile, computers get faster every year.

JDBC

Few software packages are designed without goals in mind. JDBC is one that, because of its many goals, drove the development of the API. These goals, in conjunction with early reviewer feedback, have finalized the JDBC class library into a solid framework for building database applications in Java. The goals that were set for JDBC are important. They will give you some insight as to why certain classes and functionalities behave the way they do. The eight design goals for JDBC are as follows:

SQL Level API

The designers felt that their main goal was to define a SQL interface for Java. Although not the lowest database interface level possible, it is at a low enough level for higher-level tools and APIs to be created. Conversely, it is at a high enough level for application programmers to use it confidently. Attaining this goal allows for future tool vendors to “generate” JDBC code and to hide many of JDBC’s complexities from the end user.

SQL Conformance

SQL syntax varies as you move from database vendor to database vendor. In an effort to support a wide variety of vendors, JDBC will allow any query statement to be passed through it to the underlying database driver. This allows the connectivity module to handle non-standard functionality in a manner that is suitable for its users.

JDBC must be implemented on top of common database interfaces

The JDBC SQL API must “sit” on top of other common SQL level APIs. This goal allows JDBC to use existing ODBC level drivers by the use of a software interface. This interface would translate JDBC calls to ODBC and vice versa.

Provide a Java interface that is consistent with the rest of the Java system

Because of Java’s acceptance in the user community thus far, the designers feel that they should not stray from the current design of the core Java system.

Keep it simple

This goal probably appears in all software design goal listings. JDBC is no exception. Sun felt that the design of JDBC should be very simple, allowing for only one method of completing a task per mechanism. Allowing duplicate functionality only serves to confuse the users of the API.

1. Use strong, static typing wherever possible

Strong typing allows for more error checking to be done at compile time; also, less error appear at runtime.

2. Keep the common cases simple

Because often, the usual SQL calls used by the programmer are simple SELECT's, INSERT's, DELETE's and UPDATE's, these queries should be simple to perform with JDBC. However, more complex SQL statements should also be possible.

- Finally, we decided to proceed with the implementation using Java [Networking](#).
- And for dynamically updating the cache table we go for MS [Access](#) database.
- Java has two things: a programming language and a platform.
- Java is a high-level programming language that is all the following

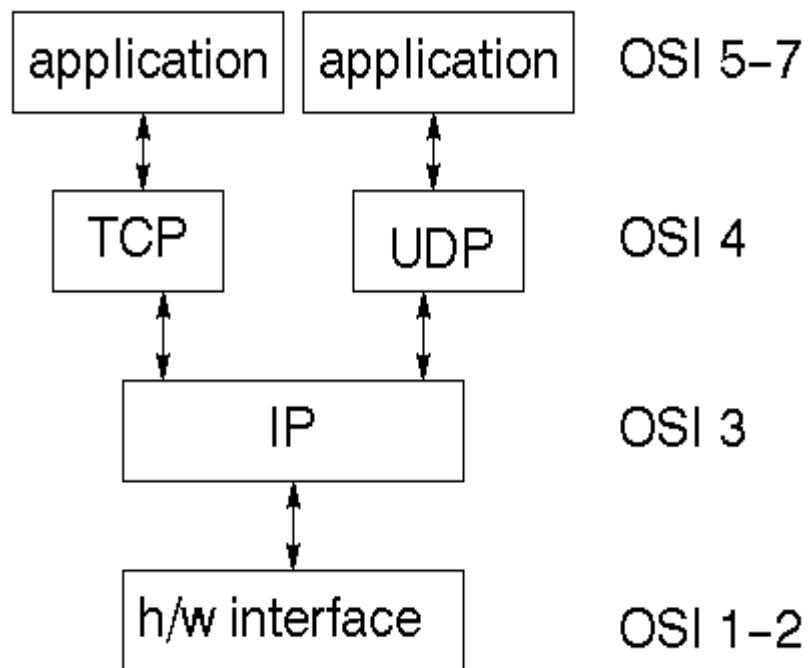
Simple	Architecture-neutral
Object-oriented	Portable
Distributed	High-performance
Interpreted	multithreaded
Robust	Dynamic
Secure	

Java is also unusual in that each Java program is both compiled and interpreted. With a compiler you translate a Java program into an intermediate language called Java byte codes. The platform-independent code instruction is passed and run on the computer. Compilation happens just once; interpretation occurs each time the program is executed. The figure illustrates how this works.

Networking

TCP/IP stack

The TCP/IP stack is shorter than the OSI one:



4.4.1. TCP block diagram

TCP is a connection-oriented protocol; UDP (User Datagram Protocol) is a connectionless protocol.

IP datagram's

The IP layer provides a connectionless and unreliable delivery system. It considers each datagram independently of the others. Any association between datagram must be supplied by the higher layers. The IP layer supplies a checksum that includes its own header. The header includes the source and destination addresses. The IP layer handles routing through an Internet. It is also responsible for breaking up large datagram into smaller ones for transmission and reassembling them at the other end.

UDP

UDP is also connectionless and unreliable. What it adds to IP is a checksum for the contents of the datagram and port numbers. These are used to give a client/server model - see later.

TCP

TCP supplies logic to give a reliable connection-oriented protocol above IP. It provides a virtual circuit that two processes can use to communicate.

Internet addresses

To use a service, you must be able to find it. The Internet uses an address scheme for machines so that they can be located. The address is a 32-bit integer which gives the IP address. This encodes a network ID and more addressing. The network ID falls into various classes according to the size of the network address.

Network address

Class A uses 8 bits for the network address with 24 bits left over for other addressing. Class B uses 16-bit network addressing. Class C uses 24-bit network addressing and class D uses all 32.

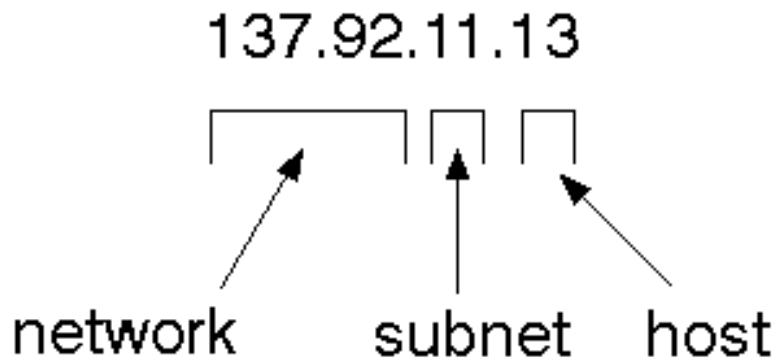
Subnet address

Internally, the UNIX network is divided into sub networks. Building 11 is currently on one sub network and uses 10-bit addressing, allowing 1024 different hosts.

Host address

8 bits are finally used for host addresses within our subnet. This places a limit of 256 machines that can be on the subnet.

Total address



The 32-bit address is usually written as 4 integers separated by dots.

Port addresses

A service exists on a host and is identified by its port. This is a 16-bit number. To send a message to a server, you send it to the port for that service of the host that it is running on. This is not location transparency! Certain of these ports are "well known".

Sockets

A socket is a data structure maintained by the system to handle network connections. A socket is created using the call `socket`. It returns an integer that is like a file descriptor. In fact, under Windows, this handle can be used with Read File and Write File functions.

```
#include <sys/types.h>#include <sys/socket.h>int socket (int family, int type, int protocol);
```

Here "family" will be `AF_INET` for IP communications, protocol will be zero, and type will depend on whether TCP or UDP is used. Two processes wishing to communicate over a network create a socket each. These are similar to the two ends of a pipe - but the actual pipe does not yet exist.

JFree Chart

JFreeChart is a free 100% Java chart library that makes it easy for developers to display professional quality charts in their applications. JFreeChart's extensive feature set includes: A consistent and well-documented API, supporting a wide range of chart types; A flexible design that is easy to extend, and targets both server-side and client-side applications; Support for many output types, including Swing components, image files (including PNG and JPEG), and vector graphics file formats (including PDF, EPS and SVG); JFreeChart is "open source" or, more specifically, [free software](#). It is distributed under the terms of the [GNU Lesser General Public Licence](#) (LGPL), which permits use in proprietary applications.

1. Map Visualizations

Charts showing values that relate to geographical areas. Some examples include: (a) population density in each state of the United States, (b) income per capita for each country in Europe, (c) life expectancy in each country of the world. The tasks in this project include: Sourcing freely redistributable vector outlines for the countries of the world, states/provinces in particular countries (USA in particular, but also other areas); Creating an appropriate dataset interface (plus default implementation), a rendered, and integrating this with the existing XYPlot class in JFreeChart;

Testing, documenting, testing some more, documenting some more.

2. Time Series Chart Interactivity

Implement a new (to JFreeChart) feature for interactive time series charts --- to display a separate control that shows a small version of ALL the time series data, with a sliding "view" rectangle that allows you to select the subset of the time series data to display in the main chart.

3. Dashboards

There is currently a lot of interest in dashboard displays. Create a flexible dashboard mechanism that supports a subset of JFreeChart chart types (dials, pies, thermometers, bars, and lines/time series) that can be delivered easily via both Java Web Start and an applet.

4. Property Editors

The property editor mechanism in JFreeChart only handles a small subset of the properties that can be set for charts. Extend (or reimplement) this mechanism to provide greater end-user control over the appearance of the charts.

J2ME (Java 2 Micro edition):

Sun Microsystems defines J2ME as "a highly optimized Java run-time environment targeting a wide range of consumer products, including pagers, cellular phones, screen-phones, digital set-top boxes and car navigation systems." Announced in June 1999 at the JavaOne Developer Conference, J2ME brings the cross-platform functionality of the Java language to smaller devices, allowing mobile wireless devices to share applications. With J2ME, Sun has adapted the Java platform for consumer products that incorporate or are based on small computing devices.

1. General J2ME architecture

J2ME uses configurations and profiles to customize the Java Runtime Environment (JRE). As a complete JRE, J2ME is comprised of a configuration, which determines the JVM used, and a profile, which defines the application by adding domain-specific classes. The configuration defines the basic run-time environment as a set of core classes and a specific JVM that run on specific types of devices. We'll discuss configurations in detail in the The profile defines the application; specifically, it adds domain-specific classes to the J2ME configuration to define certain uses for devices. We'll cover profiles in depth in the The following graphic depicts the relationship between the different virtual machines, configurations, and profiles. It also draws a parallel with the J2SE API and its Java virtual machine. While the J2SE virtual machine is generally referred to as a JVM, the J2ME virtual machines, KVM and CVM, are subsets of JVM. Both KVM and CVM can be thought of as a kind of Java virtual machine -- it's just that they are shrunken versions of the J2SE JVM and are specific to J2ME.

2.Developing J2ME applications

Introduction In this section, we will go over some considerations you need to keep in mind when developing applications for smaller devices. We'll take a look at the way the compiler is invoked when using J2SE to compile J2ME applications. Finally, we'll explore packaging and deployment and the role preverification plays in this process.

3.Design considerations for small devices

Developing applications for small devices requires you to keep certain strategies in mind during the design phase. It is best to strategically design an application for a small device before you begin coding. Correcting the code because you failed to consider all of the "gotchas" before developing the application can be a painful process. Here are some design strategies to consider:

- * Keep it simple. Remove unnecessary features, possibly making those features a separate, secondary application.
- * Smaller is better. This consideration should be a "no brainer" for all developers. Smaller applications use less memory on the device and require shorter installation times. Consider packaging your Java applications as compressed Java Archive (jar) files.
- * Minimize run-time memory use. To minimize the amount of memory used at run time, use scalar types in place of object types. Also, do not depend on the garbage collector. You should manage the memory efficiently yourself by setting object references to null when you are finished with them. Another way to reduce run-time memory is to use lazy instantiation, only allocating objects on an as-needed basis. Other ways of reducing overall and peak memory use on small devices are to release resources quickly, reuse objects, and avoid exceptions.

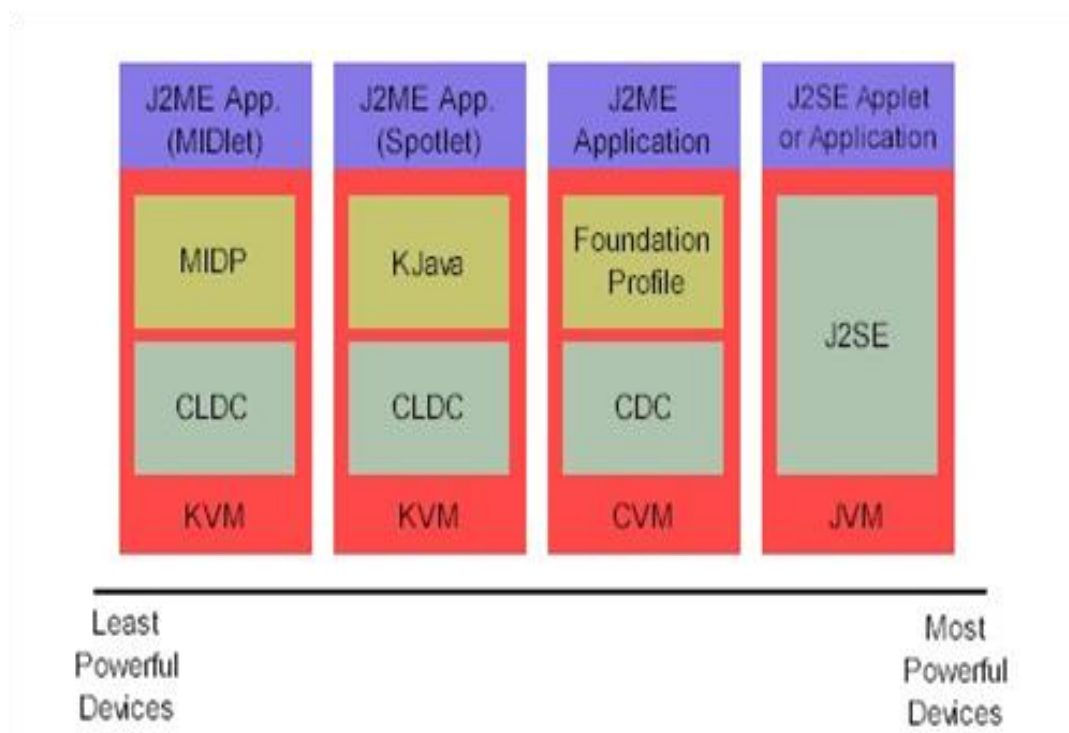
4.Configurations overview

The configuration defines the basic run-time environment as a set of core classes and a specific JVM that runs on specific types of devices. Currently, two configurations exist for J2ME, though others may be defined in the future:

Connected Limited Device Configuration (CLDC) is used specifically with the KVM for 16-bit or 32-bit devices with limited amounts of memory. This is the configuration (and the virtual machine) used for developing small J2ME applications. Its size limitations make CLDC more interesting and challenging (from a development point of view) than CDC. CLDC is also the configuration that we will use for developing our drawing tool application. An example of a small wireless device running small applications is a Palm hand-held computer.

* **Connected Device Configuration (CDC)** is used with the C virtual machine (CVM) and is used for 32-bit architectures requiring more than 2 MB of memory. An example of such a device is a Net TV box.

5.J2ME profiles



What is a J2ME profile?

As we mentioned earlier in this tutorial, a profile defines the type of device supported. The Mobile Information Device Profile (MIDP), for example, defines classes for cellular phones. It adds domain-specific classes to the J2ME configuration to define uses for similar devices. Two profiles have been defined for J2ME and are built upon CLDC: KJava and MIDP. Both KJava and MIDP are associated with CLDC and smaller devices. Profiles are built on top of configurations. Because profiles are specific to the size of the device (amount of memory) on which an application runs, certain profiles are associated with certain configurations. A skeleton profile upon which you can create your own profile, the Foundation Profile, is available for CDC.

Profile 1: KJava

KJava is Sun's proprietary profile and contains the KJava API. The KJava profile is built on top of the CLDC configuration. The KJava virtual machine, KVM, accepts the same byte codes and class file format as the classic J2SE virtual machine. KJava contains a Sun-specific API that runs on the Palm OS. The KJava API has a great deal in common with the J2SE Abstract Windowing Toolkit (AWT). However, because it is not a standard J2ME package, its main package is `com.sun.kjava`. We'll learn more about the KJava API later in this tutorial when we develop some sample applications.

Profile 2: MIDP

MIDP is geared toward mobile devices such as cellular phones and pagers. The MIDP, like KJava, is built upon CLDC and provides a standard run-time environment that allows new applications and services to be deployed dynamically on end user devices. MIDP is a common, industry-standard profile for mobile devices that is not dependent on a specific vendor. It is a complete and supported foundation for mobile application development. MIDP contains the following packages, the first three of which are core CLDC packages, plus three MIDP-specific packages.

- * `java.lang`

- * `java.io`

* java.util

* javax.microedition.io

* javax.microedition.lcdui

* javax.microedition.midlet

* javax.microedition.rm

Client Server

Overview:

With the varied topic in existence in the fields of computers, Client Server is one, which has generated more heat than light, and also more hype than reality. This technology has acquired a certain critical mass attention with its dedication conferences and magazines. Major computer vendors such as IBM and DEC, have declared that Client Servers is their main future market. A survey of DBMS magazine revealed that 76% of its readers were actively looking at the client server solution. The growth in the client server development tools from \$200 million in 1992 to more than \$1.2 billion in 1996.

Client server implementations are complex, but the underlying concept is simple and powerful. A client is an application running with local resources but able to request the database and relate the services from separate remote server. The software mediating this client server interaction is often referred to as MIDDLEWARE.

The typical client either a PC or a Workstation connected through a network to a more powerful PC, Workstation, Midrange or Main Frames server usually capable of handling request from more than one client. However, with some configuration server may also act as client. A server may need to access other server in order to process the original client request.

The key client server idea is that client as user is essentially insulated from the physical location and formats of the data needs for their application. With the proper middleware, a client input from or report can transparently access and manipulate both local database on the client machine and remote databases on one or more servers. An added bonus is the client server opens the door to multi-vendor database access indulging heterogeneous table joins.

What is a client Server

Two prominent systems in existence are client server and file server systems. It is essential to distinguish between client servers and file server systems. Both provide shared network access to data but the comparison dens there! The file server simply provides a remote disk drive that can be accessed by LAN applications on a file-by-file basis. The client server offers full relational database services such as SQL-Access, Record modifying, Insert, delete with full relational integrity backup/ restore performance for high volume of transactions, etc. the client server middleware provides a flexible interface between client and server, who does what, when and to whom.

Why Client Server

Client server has evolved to solve a problem that has been around since the earliest days of computing: how best to distribute your computing, data generation and data storage resources in order to obtain efficient, cost effective departmental and enterprise-wide data processing. During mainframe era choices were quite limited. A central machine housed both the CPU and DATA (cards, tapes, drums and later disks). Access to these resources was initially confined to batched runs that produced departmental reports at the appropriate intervals. A strong central information service department ruled the corporation. The role of the rest of the corporation limited to requesting new or more frequent reports and to provide handwritten forms from which the central data banks were created and updated. The earliest client server solutions therefore could best be characterized as "SLAVE-MASTER".

Time-sharing changed the picture. Remote terminal could view and even change the central data, subject to access permissions. And, as the central data banks evolved in to sophisticated relational database with non-programmer query languages, online users could formulate adhoc queries and produce local reports without adding to the MIS applications software backlog. However remote

access was through dumb terminals, and the client server remained subordinate to the Slave\Master.

Front end or User Interface Design

The entire user interface is planned to be developed in browser specific environment with a touch of Intranet-Based Architecture for achieving the Distributed Concept. The browser specific components are designed by using the HTML standards, and the dynamism of the designed by concentrating on the constructs of the Java Server Pages.

Communication or Database Connectivity Tier

The Communication architecture is designed by concentrating on the Standards of Servlets and Enterprise Java Beans. The database connectivity is established by using the Java Data Base Connectivity. The standards of three-tire architecture are given major concentration to keep the standards of higher cohesion and limited coupling for effectiveness of the operations.

Features of The Language Used

In my project, I have chosen Java language for developing the code.

About Java

Initially the language was called as “oak” but it was renamed as “Java” in 1995. The primary motivation of this language was the need for a platform-independent (i.e., architecture neutral) language that could be used to create software to be embedded in various consumer electronic devices.

Java is a programmer’s language.

Java is cohesive and consistent.

Except for those constraints imposed by the Internet environment, Java gives the programmer, full control.

Finally, Java is to Internet programming where C was to system programming.

Importance of Java to the Internet

Java has had a profound effect on the Internet. This is because; Java expands the Universe of objects that can move about freely in Cyberspace. In a network, two categories of objects are transmitted between the Server and the Personal computer. They are: Passive information and Dynamic active programs. The Dynamic, Self-executing programs cause serious problems in the areas of Security and probability. But, Java addresses those concerns and by doing so, has opened the door to an exciting new form of program called the Applet.

Java can be used to create two types of programs

Applications and Applets: An application is a program that runs on our Computer under the operating system of that computer. It is more or less like one creating using C or C++. Java's ability to create Applets makes it important. An Applet is an application designed to be transmitted over the Internet and executed by a Java –compatible web browser. An applet is a tiny Java program, dynamically downloaded across the network, just like an image. But the difference is, it is an intelligent program, not just a media file. It can react to the user input and dynamically change.

Features Of Java

Security

Every time you that you download a “normal” program, you are risking a viral infection. Prior to Java, most users did not download executable programs frequently, and those who did scanned them for viruses prior to execution. Most users still worried about the possibility of infecting their systems with a virus. In addition, another type of malicious program exists that must be guarded against. This type of program can gather private information, such as credit card numbers, bank account balances, and passwords. Java answers both these concerns by providing a “firewall” between a network application and your computer.

When you use a Java-compatible Web browser, you can safely download Java applets without fear of virus infection or malicious intent.

Portability

For programs to be dynamically downloaded to all the various types of platforms connected to the Internet, some means of generating portable executable code is needed. As you will see, the same mechanism that helps ensure security also helps create portability. Indeed, Java's solution to these two problems is both elegant and efficient.

The Byte code

The key that allows the Java to solve the security and portability problems is that the output of Java compiler is Byte code. Byte code is a highly optimized set of instructions designed to be executed by the Java run-time system, which is called the Java Virtual Machine (JVM). That is, in its standard form, the JVM is an interpreter for byte code.

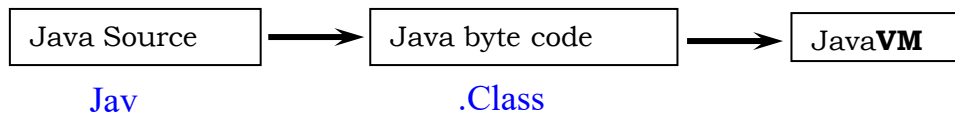
Translating a Java program into byte code helps makes it much easier to run a program in a wide variety of environments. The reason is, once the run-time package exists for a given system, any Java program can run on it.

Although Java was designed for interpretation, there is technically nothing about Java that prevents on-the-fly compilation of byte code into native code. Sun has just completed its Just In Time (JIT) compiler for byte code. When the JIT compiler is a part of JVM, it compiles byte code into executable code in real time, on a piece-by-piece, demand basis. It is not possible to compile an entire Java program into executable code all at once, because Java performs various run-time checks that can be done only at run time. The JIT compiles code, as it is needed, during execution.

Java, Virtual Machine (JVM)

Beyond the language, there is the Java virtual machine. The Java virtual machine is an important element of the Java technology. The virtual machine can be embedded within a web browser or an operating system. Once a piece of Java code is loaded onto a machine, it is verified. As part of the loading process, a class loader is invoked and does byte code verification makes sure that the code that's has been generated by the compiler will not corrupt the machine that it's loaded on. Byte code verification takes place at the end of the compilation process to make sure that is all accurate and correct. So byte code verification is integral to the compiling and executing of Java code.

Overall Description



Picture showing the development process of JAVA Program

Java programming uses to produce byte codes and executes them. The first box indicates that the Java source code is located in a .Java file that is processed with a Java compiler called javac. The Java compiler produces a file called a .class file, which contains the byte code. The .Class file is then loaded across the network or loaded locally on your machine into the execution environment is the Java virtual machine, which interprets and executes the byte code.

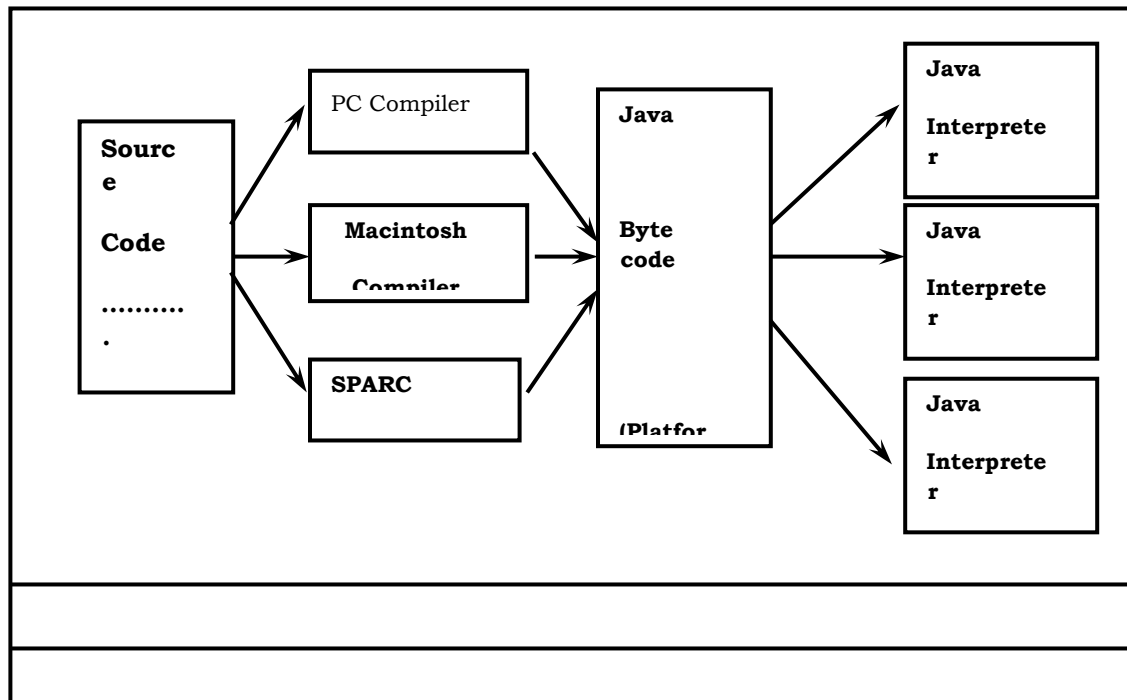
Java Architecture

Java architecture provides a portable, robust, high performing environment for development. Java provides portability by compiling the byte codes for the Java Virtual Machine, which is then interpreted on each platform by the run-time environment. Java is a dynamic system, able to load code when needed from a machine in the same room or across the planet.

Compilation of code

When you compile the code, the Java compiler creates machine code (called byte code) for a hypothetical machine called Java Virtual Machine (JVM). The JVM is supposed to execute the byte code. The JVM is created for overcoming the issue of portability. The code is written and compiled for one machine and interpreted on all machines. This machine is called Java Virtual Machine.

Compiling and interpreting Java Source Code



4.4.2. compilation and Interpretation

During run-time the Java interpreter tricks the byte code file into thinking that it is running on a Java Virtual Machine. In reality this could be a Intel Pentium Windows 95 or Sun SARC station running Solaris or Apple Macintosh running system and all could receive code from any computer through Internet and run the Applets.

Simple

Java was designed to be easy for the Professional programmer to learn and to use effectively. If you are an experienced C++ programmer, learning Java will be even easier. Because Java inherits the C/C++ syntax and many of the object oriented features of C++. Most of the confusing concepts from C++ are either left out of Java or implemented in a cleaner, more approachable manner. In Java there are a small number of clearly defined ways to accomplish a given task.

Object-Oriented

Java was not designed to be source-code compatible with any other language. This allowed the Java team the freedom to design with a blank slate. One outcome of this was a clean usable, pragmatic approach to objects. The object model in Java is simple and easy to extend, while simple types, such as integers, are kept as high-performance non-objects.

Robust

The multi-platform environment of the Web places extraordinary demands on a program, because the program must execute reliably in a variety of systems. The ability to create robust programs was given a high priority in the design of Java. Java is strictly typed language; it checks your code at compile time and run time. Java virtually eliminates the problems of memory management and deallocation, which is completely automatic. In a well-written Java program, all run time errors can –and should –be managed by your program.

Java Script

JavaScript is a script-based programming language that was developed by Netscape Communication Corporation. JavaScript was originally called Live Script and renamed as JavaScript to indicate its relationship with Java. JavaScript supports the development of both client and server components of Web-based applications. On the client side, it can be used to write programs that are executed by a Web browser within the context of a Web page. On the server side, it can be used to write Web server programs that can process information submitted by a Web browser and then updates the browser's display accordingly. Even though JavaScript supports both client and server Web programming, we prefer JavaScript at Client-side programming since most of the browsers supports it. JavaScript is almost as easy to learn as HTML, and JavaScript statements can be included in HTML documents by enclosing the statements between a pair of scripting tags

```
<SCRIPTS> ..</SCRIPT>.
```

```
<SCRIPT LANGUAGE = "JavaScript">
```

JavaScript statements

```
</SCRIPT>
```


Here are a few things we can do with JavaScript:

1. Validate the contents of a form and make calculations.
2. Add scrolling or changing messages to the Browser's status line.
3. Animate images or rotate images that change when we move the mouse over them.
4. Detect the browser in use and display different content for different browsers.
5. Detect installed plug-ins and notify the user if a plug-in is required.
6. We can do much more with JavaScript, including creating entire application.

JavaScript Vs Java

JavaScript and Java are entirely different languages. A few of the most glaring differences are:

Java applets are generally displayed in a box within the web document; JavaScript can affect any part of the Web document itself.

While JavaScript is best suited to simple applications and adding interactive features to Web pages; Java can be used for incredibly complex applications.

There are many other differences but the important thing to remember is that JavaScript and Java are separate languages. They are both useful for different things; in fact they can be used together to combine their advantages.

ADVANTAGES

1. JavaScript can be used for Sever-side and Client-side scripting.
2. It is more flexible than VBScript.
3. JavaScript is the default scripting languages at Client-side since all the browsers supports it.

Hyper Text Markup Language

Hypertext Markup Language (HTML), the languages of the World Wide Web (WWW), allows users to produces Web pages that include text, graphics and pointer to other Web pages (Hyperlinks).

HTML is not a programming language but it is an application of ISO Standard 8879, SGML (Standard Generalized Markup Language), but specialized to hypertext and adapted to the Web. The idea behind Hypertext is that instead of reading text in rigid linear structure, we can easily jump from one point to another point. We can navigate through the information based on our interest and preference. A markup language is simply a series of elements, each delimited with special characters that define how text or other items enclosed within the elements should be displayed. Hyperlinks are underlined or emphasized works that load to other documents or some portions of the same document.

HTML can be used to display any type of document on the host computer, which can be geographically at a different location. It is a versatile language and can be used on any platform or desktop.

HTML provides tags (special codes) to make the document look attractive. HTML tags are not case-sensitive. Using graphics, fonts, different sizes, color, etc., can enhance the presentation of the document. Anything that is not a tag is part of the document itself.

Basic HTML Tags :

<code><!-- --></code>	Specifies comments
<code><A>.....</code>	Creates hypertext links
<code>.....</code>	Formats text as bold
<code><BIG>.....</BIG></code>	Formats text in large font.
<code><BODY>...</BODY></code>	Contains all tags and text in the HTML document
<code><CENTER>...</CENTER></code>	Creates text
<code><DD>...</DD></code>	Definition of a term
<code><DL>...</DL></code>	Creates definition list
<code>...</code>	Formats text with a particular font

<FORM>...</FORM>	Encloses a fill-out form
<FRAME>...</FRAME>	Defines a particular frame in a set of frames
<H#>...</H#>	Creates headings of different levels
<HEAD>...</HEAD>	Contains tags that specify information about a document
<HR>...</HR>	Creates a horizontal rule
<HTML>...</HTML>	Contains all other HTML tags
<META>...</META>	Provides meta-information about a document
<SCRIPT>...</SCRIPT>	Contains client-side or server-side script
<TABLE>...</TABLE>	Creates a table
<TD>...</TD>	Indicates table data in a table
<TR>...</TR>	Designates a table row
<TH>...</TH>	Creates a heading in a table

ADVANTAGES

A HTML document is small and hence easy to send over the net. It is small because it does not include formatted information.

HTML is platform independent.

HTML tags are not case-sensitive.

Java Database Connectivity

What Is JDBC?

JDBC is a Java API for executing SQL statements. (As a point of interest, JDBC is a trademarked name and is not an acronym; nevertheless, JDBC is often thought of as standing for Java Database

Connectivity. It consists of a set of classes and interfaces written in the Java programming language. JDBC provides a standard API for tool/database developers and makes it possible to write database applications using a pure Java API.

Using JDBC, it is easy to send SQL statements to virtually any relational database. One can write a single program using the JDBC API, and the program will be able to send SQL statements to the appropriate database. The combinations of Java and JDBC lets a programmer write it once and run it anywhere.

What Does JDBC Do?

Simply put, JDBC makes it possible to do three things:

- Establish a connection with a database
- Send SQL statements
- Process the results.

JDBC versus ODBC and other APIs

At this point, Microsoft's ODBC (Open Database Connectivity) API is that probably the most widely used programming interface for accessing relational databases. It offers the ability to connect to almost all databases on almost all platforms.

So why not just use ODBC from Java? The answer is that you can use ODBC from Java, but this is best done with the help of JDBC in the form of the JDBC-ODBC Bridge, which we will cover shortly. The question now becomes "Why do you need JDBC?" There are several answers to this question:

ODBC is not appropriate for direct use from Java because it uses a C interface. Calls from Java to native C code have a number of drawbacks in the security, implementation, robustness, and automatic portability of applications. A literal translation of the ODBC C API into a Java API would not be desirable. For example, Java has no pointers, and ODBC makes copious use of them, including the notoriously error-prone generic pointer "void *". You can think of JDBC as ODBC translated into an object-oriented interface that is natural for Java programmers.

ODBC is hard to learn. It mixes simple and advanced features together, and it has complex options even for simple queries. JDBC, on the other hand, was designed to keep simple things simple while allowing more advanced capabilities where required. A Java API like JDBC is needed in order to enable a "pure Java" solution. When ODBC is used, the ODBC driver manager and drivers must be manually installed on every client machine. When the JDBC driver is written completely in Java, however, JDBC code is automatically installable, portable, and secure on all Java platforms from network computers to mainframes.

Two-tier and Three-tier Models

The JDBC API supports both two-tier and three-tier models for database access.

In the two-tier model, a Java applet or application talks directly to the database. This requires a JDBC driver that can communicate with the particular database management system being accessed. A user's SQL statements are delivered to the database, and the results of those statements are sent back to the user. The database may be located on another machine to which the user is connected via a network. This is referred to as a client/server configuration, with the user's machine as the client, and the machine housing the database as the server. The network can be an Intranet, which, for example, connects employees within a corporation, or it can be the Internet.

In the three-tier model, commands are sent to a "middle tier" of services, which then send SQL statements to the database. The database processes the SQL statements and sends the results back to the middle tier, which then sends them to the user. MIS directors find the three-tier model very attractive because the middle tier makes it possible to maintain control over access and the kinds of updates that can be made to corporate data. Another advantage is that when there is a middle tier, the user can employ an easy-to-use higher-level API which is translated by the middle tier into the appropriate low-level calls. Finally, in many cases the three-tier architecture can provide performance advantages.

Until now the middle tier has typically been written in languages such as C or C++, which offer fast performance. However, with the introduction of optimizing compilers that translate Java byte code into efficient machine-specific code, it is becoming practical to implement the middle tier in Java. This is a big plus, making it possible to take advantage of Java's robustness, multithreading, and security features. JDBC is important to allow database access from a Java middle tier.

JDBC Driver Types

The JDBC drivers that we are aware of at this time fit into one of four categories:

- JDBC-ODBC bridge plus ODBC driver
- Native-API partly-Java driver
- JDBC-Net pure Java driver
- Native-protocol pure Java driver
- JDBC-ODBC Bridge

If possible, use a Pure Java JDBC driver instead of the Bridge and an ODBC driver. This eliminates the client configuration required by ODBC. It also eliminates the potential that the Java VM could be corrupted by an error in the native code brought in by the Bridge (that is, the Bridge native library, the ODBC driver manager library, the ODBC driver library, and the database client library).

What is the JDBC- ODBC Bridge?

The JDBC-ODBC Bridge is a JDBC driver, which implements JDBC operations by translating them into ODBC operations. To ODBC it appears as a normal application program. The Bridge implements JDBC for any database for which an ODBC driver is available. The Bridge is implemented as the

`sun.jdbc.odbc` Java package and contains a native library used to access ODBC. The Bridge is a joint development of Intersolv and JavaSoft.

Java Server Pages (JSP)

Java server Pages is a simple, yet powerful technology for creating and maintaining dynamic-content web pages. Based on the Java programming language, Java Server Pages offers proven portability, open standards, and a mature re-usable component model. The Java Server Pages architecture enables the separation of content generation from content presentation. This separation does not ease maintenance headaches, it also allows web team members to focus on their areas of expertise. Now, web page designer can concentrate on layout, and web application designers on programming, with minimal concern about impacting each other's work.

Features of JSP

Portability:

Java Server Pages files can be run on any web server or web-enabled application server that provides support for them. Dubbed the JSP engine, this support involves recognition, translation, and management of the Java Server Page lifecycle and its interaction components.

Components

It was mentioned earlier that the Java Server Pages architecture can include reusable Java components. The architecture also allows for the embedding of a scripting language directly into the Java Server Pages file. The components currently supported include Java Beans, and Servlets.

Processing

A Java Server Pages file is essentially an HTML document with JSP scripting or tags. The Java Server Pages file has a JSP extension to the server as a Java Server Pages file. Before the page is served, the Java Server Pages syntax is parsed and processed into a Servlet on the server side. The Servlet that is generated outputs real content in straight HTML for responding to the client.

Access Models:

A Java Server Pages file may be accessed in at least two different ways. A client's request comes directly into a Java Server Page. In this scenario, suppose the page accesses reusable Java Bean components that perform particular well-defined computations like accessing a database. The result of the Beans computations, called result sets is stored within the Bean as properties. The page uses such Beans to generate dynamic content and present it back to the client. In both of the above cases, the page could also contain any valid Java code. Java Server Pages architecture encourages separation of content from presentation.

Steps in the execution of a JSP Application:

The client sends a request to the web server for a JSP file by giving the name of the JSP file within the form tag of a HTML page. This request is transferred to the JavaWebServer. At the server side JavaWebServer receives the request and if it is a request for a jsp file server gives this request to the JSP engine. JSP engine is program which can understand the tags of the jsp and then it converts those tags into a Servlet program and it is stored at the server side. This Servlet is loaded in the memory and then it is executed and the result is given back to the JavaWebServer and then it is transferred back to the client.

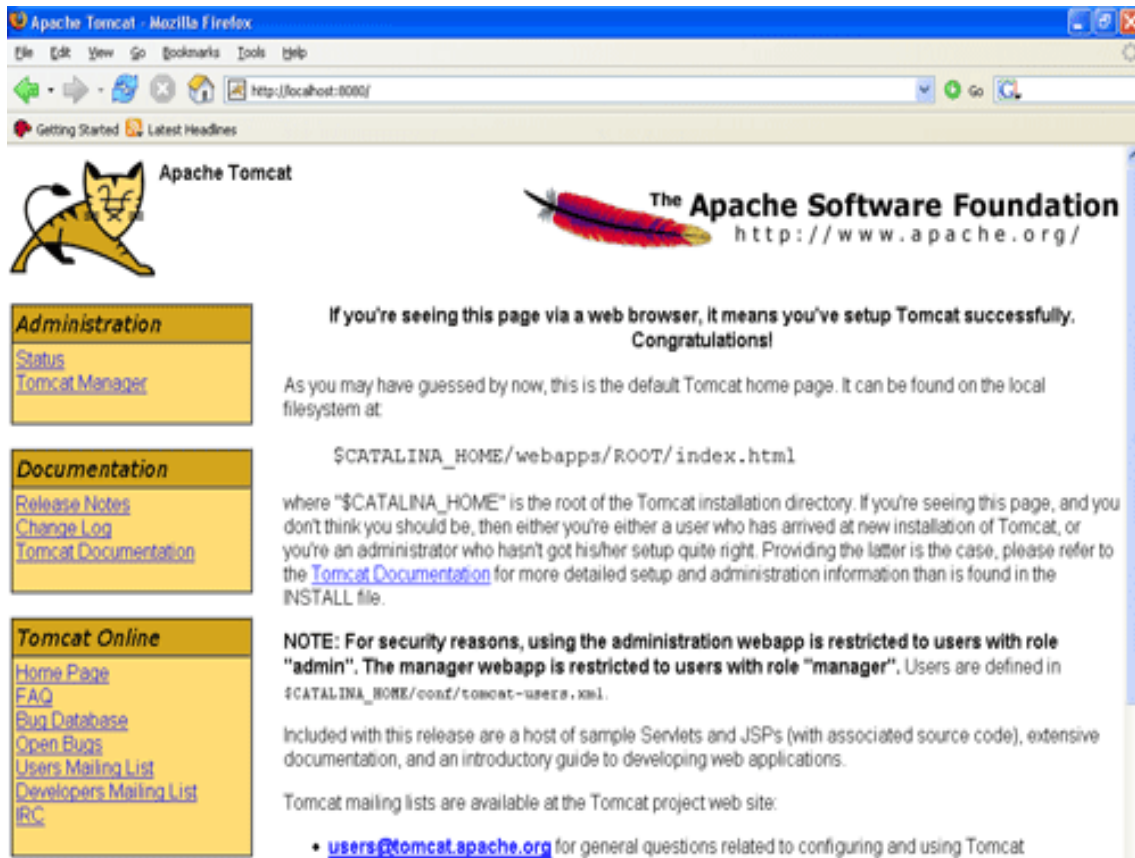
JDBC connectivity

The JDBC provides database-independent connectivity between the J2EE platform and a wide range of tabular data sources. JDBC technology allows an Application Component Provider to:

- Perform connection and authentication to a database server
- Manage transactions
- Move SQL statements to a database engine for preprocessing and execution
- Execute stored procedures
- Inspect and modify the results from Select statements.

Tomcat 6.0 web server

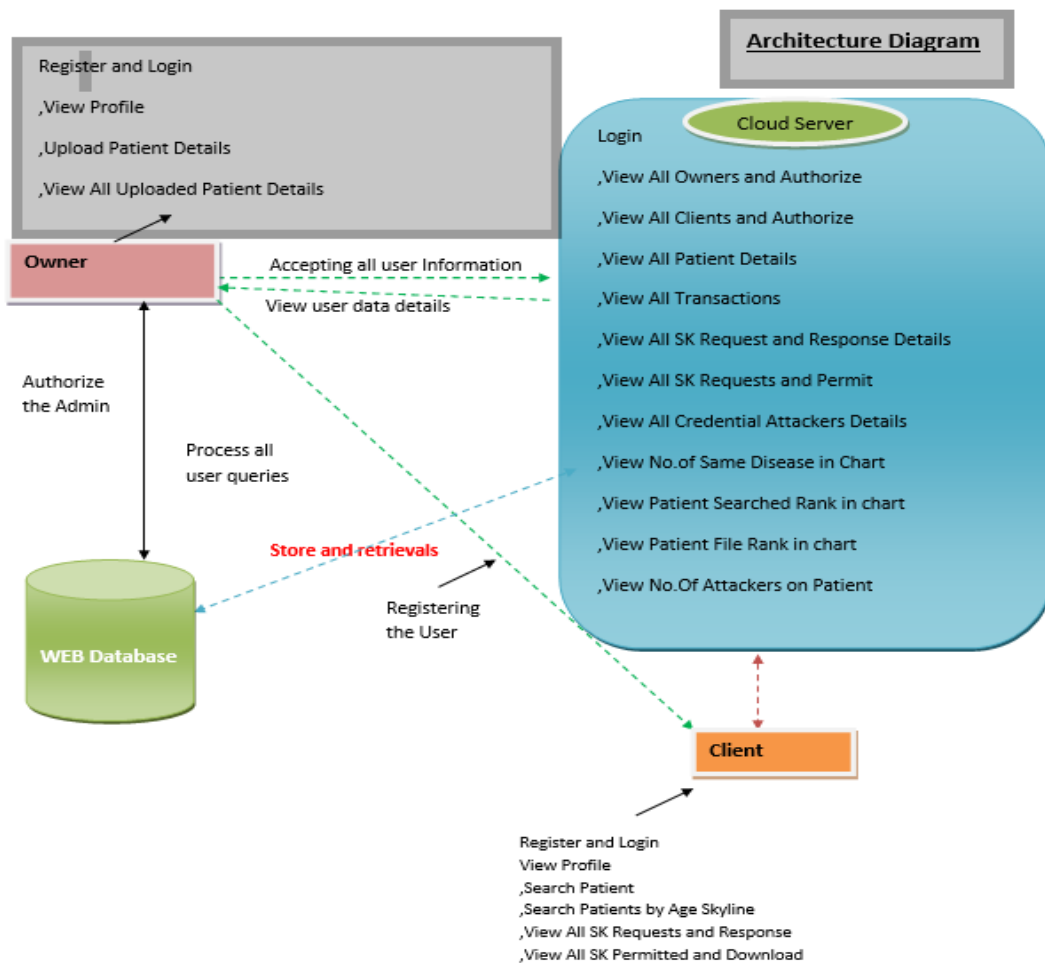
Tomcat is an open source web server developed by Apache Group. Apache Tomcat is the servlet container that is used in the official Reference Implementation for the Java Servlet and Java Server Pages technologies. The Java Servlet and Java Server Pages specifications are developed by Sun under the Java Community Process. Web Servers like Apache Tomcat support only web components while an application server supports web components as well as business components (BEAs Weblogic, is one of the popular application server). To develop a web application with jsp/servlet install any web server like JRun, Tomcat etc to run your application.



4.4.1. Tomcat web server

5. SOFTWARE DESIGN

5.1 System Architecture:



5.1 System Architecture

5.2 Data Flow Diagram:

DFD SYMBOLS:

In the DFD, there are four symbols

1. A square defines a source(originator) or destination of system data
2. An arrow identifies data flow. It is the pipeline through which the information flows
3. A circle or a bubble represents a process that transforms incoming data flow into outgoing data flows.
4. An open rectangle is a data store, data at rest or a temporary repository of data.

CONSTRUCTING A DFD:

Several rules of thumb are used in drawing DFD'S:

1. Process should be named and numbered for an easy reference. Each name should be representative of the process.
2. The direction of flow is from top to bottom and from left to right. Data traditionally flow from source to the destination although they may flow back to the source. One way to indicate this is to draw long flow line back to a source. An alternative way is to repeat the source symbol as a destination. Since it is used more than once in the DFD it is marked with a short diagonal.
3. When a process is exploded into lower level details, they are numbered.
4. The names of data stores and destinations are written in capital letters. Process and dataflow names have the first letter of each word capitalized

A DFD typically shows the minimum contents of data store. Each data store should contain all the data elements that flow in and out.

Questionnaires should contain all the data elements that flow in and out. Missing interfaces redundancies and like is then accounted for often through interviews.

SAILENT FEATURES OF DFD'S:

1. The DFD shows flow of data, not of control loops and decision are controlled considerations do not appear on a DFD.
2. The DFD does not indicate the time factor involved in any process whether the dataflow take place daily, weekly, monthly or yearly.
3. The sequence of events is not brought out on the DFD.

TYPES OF DATA FLOW DIAGRAMS:

1. Current Physical
2. Current Logical
3. New Logical
4. New Physical

CURRENT PHYSICAL:

In Current Physical DFD process label include the name of people or their positions or the names of computer systems that might provide some of the overall system-processing label includes an identification of the technology used to process the data. Similarly data flows and data stores are often labels with the names of the actual physical media on which data are stored such as file folders, computer files, business forms or computer tapes.

CURRENT LOGICAL:

The physical aspects at the system are removed as much as possible so that the current system is reduced to its essence to the data and the processors that transforms them regardless of actual physical form.

NEW LOGICAL:

This is exactly like a current logical model if the user were completely happy with the user were completely happy with the functionality of the current system but had problems with how it was implemented typically through the new logical model will differ from current logical model while having additional functions, absolute function removal and inefficient flows recognized.

NEW PHYSICAL:

The new physical represents only the physical implementation of the new system.

RULES GOVERNING THE DFD'S:

PROCESS:

1. No process can have only outputs.
2. No process can have only inputs. If an object has only inputs than it must be a sink.
3. A process has a verb phrase label.

DATA STORE:

1. Data cannot move directly from one data store to another data store, a process must move data.
2. Data cannot move directly from an outside source to a data store, a process, which receives, must move data from the source and place the data into data store
3. A data store has a noun phrase label.

SOURCE OR SINK:

The origin and / or destination of data.

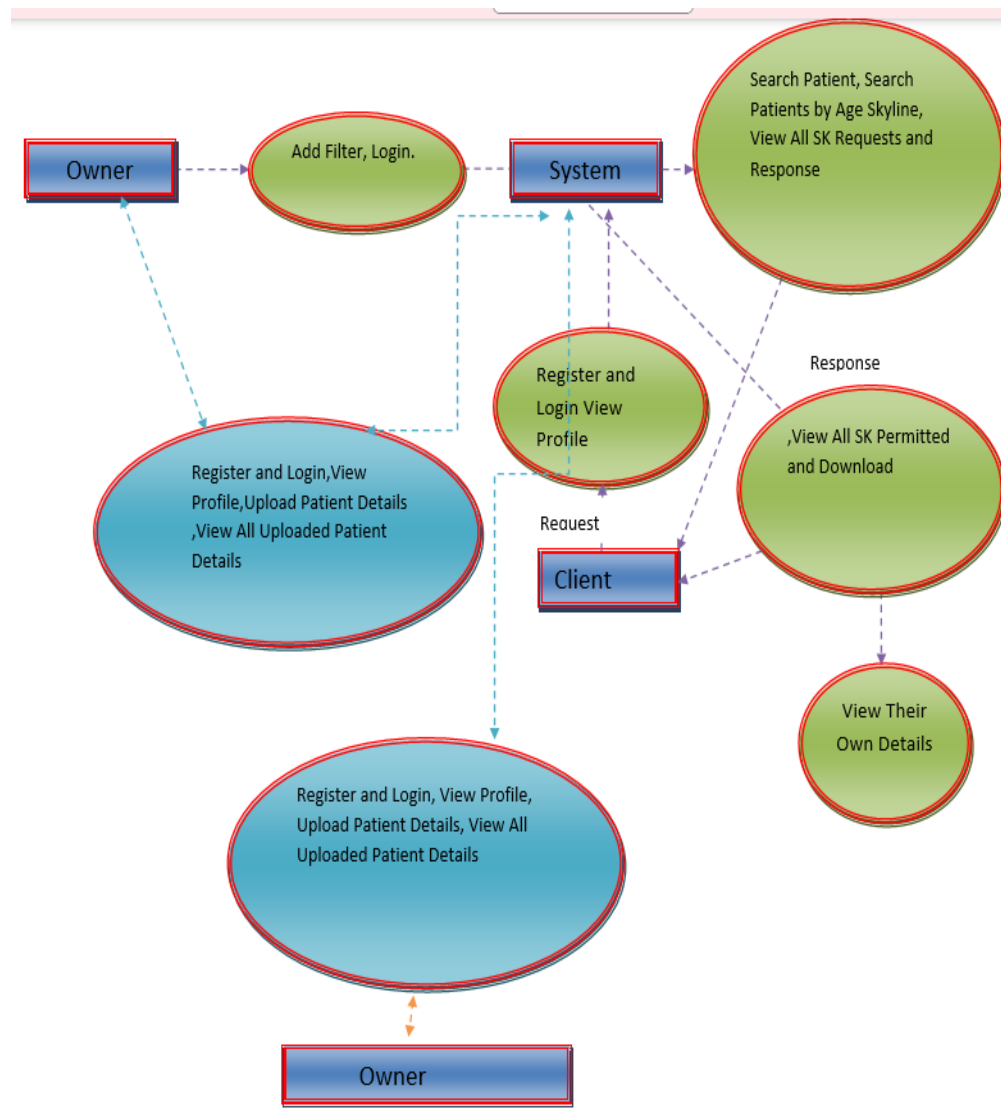
1. Data cannot move direly from a source to sink it must be moved by a process
2. A source and /or sink has a noun phrase land

DATA FLOW:

1. A Data Flow has only one direction of flow between symbols. It may flow in both directions between a process and a data store to show a read before an update. The later is usually indicated however by two separate arrows since these happen at different type.
2. A join in DFD means that exactly the same data comes from any of two or more different processes data store or sink to a common location.
3. A data flow cannot go directly back to the same process it leads. There must be atleast one other process that handles the data flow produce some other data flow returns the original data into the beginning process.
4. A Data flow to a data store means update (delete or change).
5. A data Flow from a data store means retrieve or use.

A data flow has a noun phrase label more than one data flow noun phrase can appear on a single arrow as long as all of the flows on the same arrow move together as one package.

DFD Diagrams:



5.1.1 DFD diagram

5.3 UML Diagrams:

The Unified Modeling Language allows the software engineer to express an analysis model using the modeling notation that is governed by a set of syntactic semantic and pragmatic rules. A UML system is represented using five different views that describe the system from distinctly different perspective. Each view is defined by a set of diagram, which is as follows.

1. □ User Model View
 - i. This view represents the system from the users perspective.
 - ii. The analysis representation describes a usage scenario from the end-users perspective.
2. □ Structural Model view
 - i. In this model the data and functionality are arrived from inside the system.
 - ii. This model view models the static structures.

- Behavioral Model View

It represents the dynamic of behavioral as parts of the system, depicting the interactions of collection between various structural elements described in the user model and structural model view.

- Implementation Model View

In this the structural and behavioral as parts of the system are represented as they are to be built.

- Environmental Model View

In this the structural and behavioral aspects of the environment in which the system is to be implemented are represented.

Class diagram:-

The class diagram is the main building block of object oriented modeling. It is used both for general conceptual modeling of the systematic of the application, and for detailed modeling translating the models into programming code. Class diagrams can also be used for data modeling. The classes in

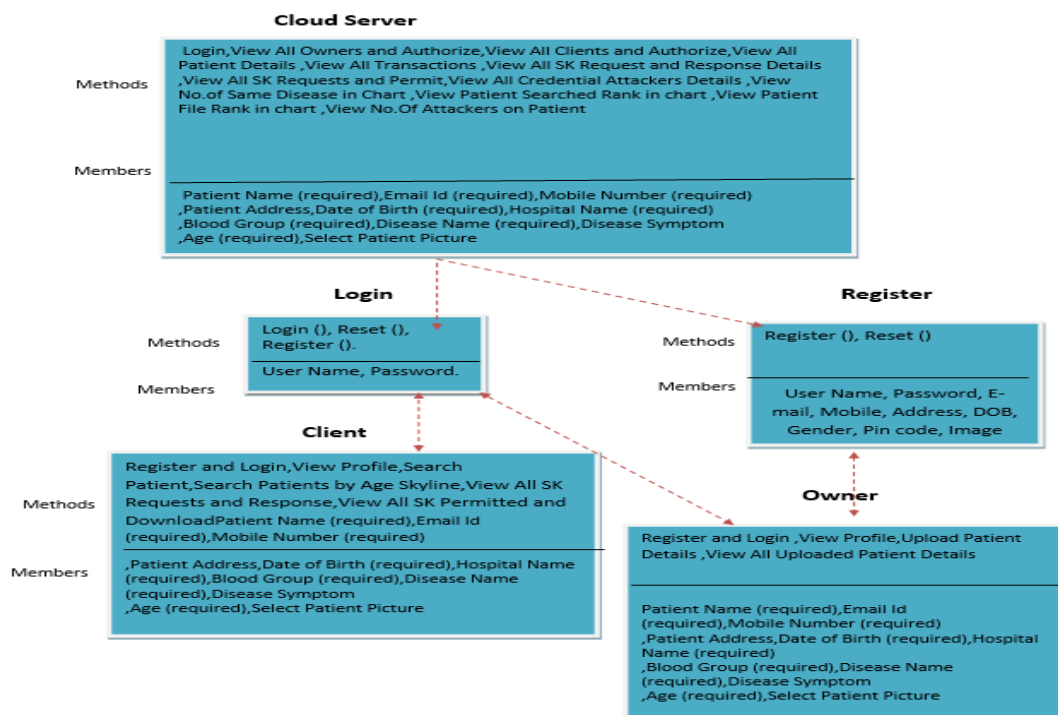
a class diagram represent both the main objects, interactions in the application and the classes to be programmed.

A class with three sections

In the diagram, classes are represented with boxes which contain three parts:

- ☐ The upper part holds the name of the class
- ☐ The middle part contains the attributes of the class
- ☐ The bottom part gives the methods or operations the class can take or undertake

Class Diagram:

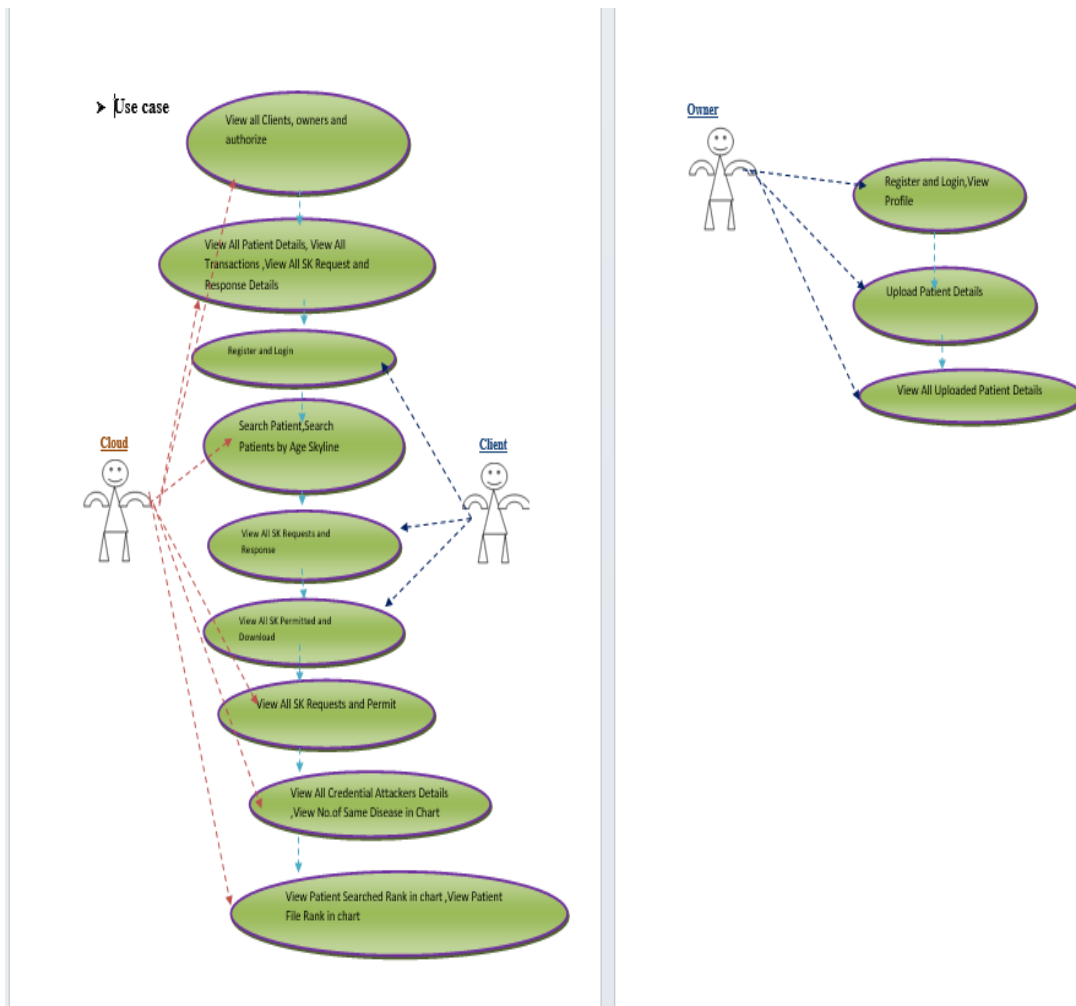


5.2.1 Class diagram

Use case diagram: -

A use case diagram at its simplest is a representation of a user's interaction with the system and depicting the specifications of a use case. A use case diagram can portray the different types of users of a system and the various ways that they interact with the system. This type of diagram is typically used in conjunction with the textual use case and will often be accompanied by other types of diagrams as well.

Use case diagram:

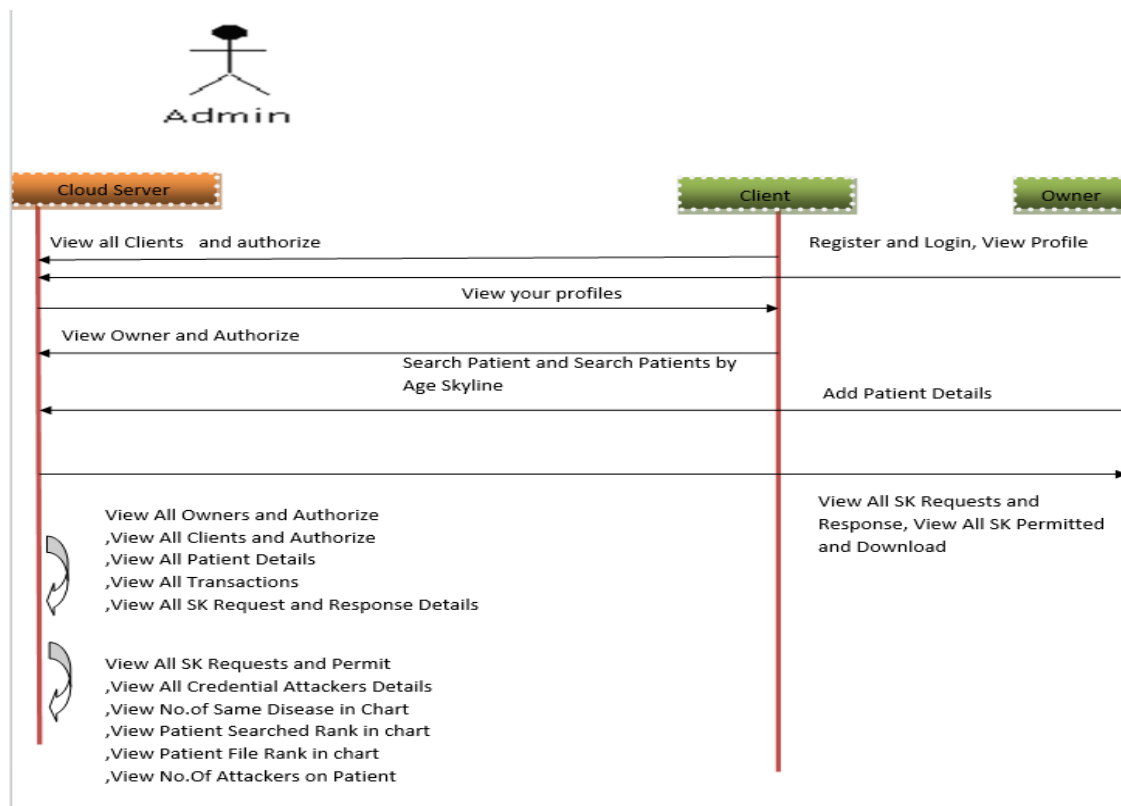


5.2.2. Use case diagram

Sequence Diagram: -

A sequence diagram is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. A sequence diagram shows object interactions arranged in time sequence. It depicts the objects and classes involved in the scenario and the sequence of messages exchanged between the objects needed to carry out the functionality of the scenario. Sequence diagrams are typically associated with use case realizations in the Logical View of the system under development. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

Sequence diagram:



5.2.3. Sequence Diagram

6. CODING AND IMPLEMENTATION

6.1. Sample code

```
<?xml version="1.0" encoding="UTF-8">

<web-app
xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance"
xmlns="http://java.sun.com/xml/ns/javaee"
xmlns:web="http://java.sun.com/xml/ns/javaee/web-
app_2_5.xsd"
xsi:schemaLocation="http://java.sun.com/xml/ns/ja
vae http://java.sun.com/xml/ns/javaee/web-
app_2_5.xsd" id="WebApp_ID" version="2.5">

  <display-name>SecureandEfficient Sky line
queries</display-name>

  <welcome-file-list>

    <welcome-file>index.html</welcome-file>

    <welcome-file>index.htm</welcome-file>

    <welcome-file>index.jsp</welcome-file>

    <welcome-file>default.html</welcome-file>

    <welcome-file>default.htm</welcome-file>

    <welcome-file>default.jsp</welcome-file>

  </welcome-file-list>

</web-app>

<%@ page import="java.sql.*"%>
<%@ page import="java.util.*" %>
<%
Connection connection = null;
try {
Class.forName("com.mysql.jdbc.Driver");
```

```

        connection =
DriverManager.getConnection("jdbc:mysql://localhost:3306/Efficient_Skyline","root","");
        String sql="";
    }
catch(Exception e)
{
    System.out.println(e);
}
%>

```

```

<!DOCTYPE html PUBLIC "-//W3C//DTD
XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-
transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<title> Patients Access Control Details</title>
<meta http-equiv="Content-Type"
content="text/html; charset=utf-8" />
<link href="css/style.css" rel="stylesheet"
type="text/css" />
<link rel="stylesheet" type="text/css"
href="css/coin-slider.css" />
<script type="text/javascript" src="js/cufon-
yui.js"></script>
<script type="text/javascript" src="js/cufon-
aller.js"></script>
<script type="text/javascript" src="js/jquery-
1.4.2.min.js"></script>
<script type="text/javascript"
src="js/script.js"></script>
<script type="text/javascript" src="js/coin-
slider.min.js"></script>

```

```

<script language="javascript"
type="text/javascript">
</script>
<style type="text/css">
<!--
.style1 {
font-size: 25px;
color: #33FF99;
}
.style2 {font-size: 25px}
.style16 {font-size: 15px; }
.style17 {font-size: 12px; }
.style18 {color: #FFFFFF}
.style19 {font-size: 14px}
-->
</style>
</head>
<body>
<div class="main">
  <div class="header">
    <div class="header_resize">
      <div class="logo">
        <h1><a href="index.html" class="style1
style2">Secure and Efficient Skyline Queries on
Encrypted Data <br />
</a></h1>
      </div>
      <div class="menu_nav">
        <ul>
          <li><a href="index.html"><span>Home
Page</span></a></li>

```

```
        <li><a
href="owner_login.jsp"><span>Owner</span></a>
</li>
```

```
        <li><a
href="user_login.jsp"><span>Client</span></a></li>
```

```
        <li class="active"><a
href="cs_login.jsp"><span>Cloud
Server</span></a></li>
```

```
        <li><a
href="au_login.jsp"><span></span></a></li>
```

```
    </ul>
```

```
</div>
```

```
<div class="clr"></div>
```

```
<div class="slider">
```

```
    <div id="coin-slider"> <a href="#"> </a> <a href="#"> </a> <a href="#"> </a> </div>
```

```
    <div class="clr"></div>
```

```
</div>
```

```
<div class="clr"></div>
```

```
</div>
```

```
</div>
```

```
<div class="content">
```

```
    <div class="content_resize">
```

```
        <div class="mainbar">
```

```
            <div class="article">
```

```
                <h2 align="center">View All Access Control
Provided Patients and Details</h2>
```

```
                <p align="center">&nbsp;</p>
```

```

<table width="704" border="1" align="center"
cellspacing="0" cellpadding="5">

  <tr>

    <td width="79" bgcolor="#FF0000"><div
align="center" class="style18
style19"><strong>Id</strong></div></td>

    <td width="175" bgcolor="#FF0000"><div
align="center" class="style18
style19"><strong>Patient Name
</strong></div></td>

    <td width="155" bgcolor="#FF0000"><div
align="center" class="style18
style19"><strong>Department</strong></div></td>

    <td width="141" bgcolor="#FF0000"><div
align="center" class="style18
style19"><strong>Profession</strong></div></td>

    <td width="92" bgcolor="#FF0000"><div
align="center" class="style19
style18"><strong>Date</strong></div></td>

  </tr>

<%@page
import="java.io.BufferedInputStream"%>

<%@page
import="java.security.DigestInputStream"%>

<%@page import="java.io.FileInputStream"%>

<%@page import="java.io.PrintStream"%>

<%@page import="java.io.FileOutputStream"%>

<%@page import="java.math.BigInteger"%>

<%@ page
import="java.security.Key,java.security.KeyPair,jav
a.security.KeyPairGenerator,javax.crypto.Cipher"%
>

<%@ include file="connect.jsp"%>

<%@page
import="java.util.*,java.security.Key,java.util.Rand
om,javax.crypto.Cipher,javax.crypto.spec.SecretKe
ySpec,org.bouncycastle.util.encoders.Base64"

```

```

<%@page
import="java.security.MessageDigest"%>
<%@page import="java.sql.Statement"%>
<%@page import="java.sql.ResultSet"%>
<%@page
import="java.text.SimpleDateFormat"%>
<%@page import="java.util.Date"%>

<%
String s1 = "", s2 = "", s3 = "", s4 = "", s5 = "", s6 =
"", s7 = "", s8, s9 = "", s10, s11, s12,
s13,s14,s15,s16,s17;
int i = 0, j = 0, k = 0;

    String
user=(String)application.getAttribute("ow");

try {

String query = "select * from acc_control_per ";
Statement st = connection.createStatement();
ResultSet rs = st.executeQuery(query);
while (rs.next()) {
i = rs.getInt(1);
s2 = rs.getString(2);
s3 = rs.getString(3);
s4 = rs.getString(4);
s5 = rs.getString(5);
s6 = rs.getString(6);
%> <tr>
<td height="29">
<div align="center"
class="style16"><%=i%></div>
</td>
<td height="29">

```



```

<div align="center"
class="style16"><%=s2%></div>          </td>
<td>

<div align="center"
class="style16"><%=s3%></div>          </td>
<td>

<div align="center"
class="style16"><%=s4%></div>          </td>
<td>

<div align="center"
class="style16"><%=s5%></div>          </td>

</tr>

<%
}

connection.close();
}

catch (Exception e)
{
out.println(e.getMessage());
}
%>
</table>

<p>&nbsp;</p>

<p align="center"><a href="s_main.jsp"
class="style17">Back</a></p>

<p>&nbsp;</p>

</div>

</div>

<div class="sidebar">

```

```

    <div class="gadget">
        <h2 class="star"><span>Server</span>
Menu</h2>

        <div class="clr"></div>

        <ul class="sb_menu">
            <li><a href="s_main.jsp"><span>Server
Main</span></a></li>
            <li><a href="cs_login.jsp"><span>Log
Out</span></a></li>
        </ul>
    </div>
</div>

<div class="clr"></div>
</div>

<div class="fbg"></div>
<div class="footer">
    <div class="footer_resize">
        <div style="clear:both;"></div>
    </div>
</div>
</div>
<div align=center></div>
</body>
</html>

```

```

<!DOCTYPE html PUBLIC "-//W3C//DTD
XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-
transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head>

```

```

<title>View Attackers</title>

<meta http-equiv="Content-Type"
content="text/html; charset=utf-8" />

<link href="css/style.css" rel="stylesheet"
type="text/css" />

<link rel="stylesheet" type="text/css"
href="css/coin-slider.css" />

<script type="text/javascript" src="js/cufon-
yui.js"></script>

<script type="text/javascript" src="js/cufon-
aller.js"></script>

<script type="text/javascript" src="js/jquery-
1.4.2.min.js"></script>

<script type="text/javascript"
src="js/script.js"></script>

<script type="text/javascript" src="js/coin-
slider.min.js"></script>

<script language="javascript"
type="text/javascript">
</script>

<style type="text/css">

<!--

.style1 {
font-size: 25px;
color: #33FF99;
}

.style2 {font-size: 25px}
.style4 {font-size: 16px}
.style5 {color: #FFFFFF}
.style6 {font-size: 14px}
.style7 {color: #FF0000}
.style9 {font-weight: bold}
.style11 {font-weight: bold}
.style13 {font-weight: bold}

```

```

.style14 {font-weight: bold}
.style15 {font-weight: bold}
.style16 {color: #FFFF00}
-->
</style>
</head>
<body>
<div class="main">
  <div class="header">
    <div class="header_resize">
      <div class="logo">
        <h1><a href="index.html" class="style1
style2">Secure and Efficient Skyline Queries on
Encrypted Data <br />
</a></h1>
      </div>
      <div class="menu_nav">
        <ul>
          <li><a href="index.html"><span>Home
Page</span></a></li>
          <li><a
href="owner_login.jsp"><span>Owner</span></a>
</li>
          <li><a
href="user_login.jsp"><span>Client</span></a></li>
          <li class="active"><a
href="cs_login.jsp"><span>Cloud
Server</span></a></li>
          <li><a
href="au_login.jsp"><span></span></a></li>
        </ul>
      </div>
      <div class="clr"></div>

```

```

<div class="slider">
    <div id="coin-slider"> <a href="#"> </a> <a href="#"> </a> <a href="#"> </a> </div>
    <div class="clr"></div>
</div>
<div class="clr"></div>
</div>
<div class="content">
    <div class="content_resize">
        <div class="mainbar">
            <div class="article">
                <h2 align="center" class="style4"> View All
Attacker Details [ <a href="cs_chart4.jsp">View in
Chart</a> ]</h2>
                <p align="center">&nbsp;</p>
                <table width="691" border="1"
align="center" cellpadding="0" cellspacing="0" ">
                    <tr>
                        <td width="31" height="46"
valign="middle" bgcolor="#FF0000" style="color:
#2c83b0;"><div align="center" class="style7
style21 style15 style5
style6"><strong>ID</strong></div></td>
                        <td width="181" height="46" valign="middle"
bgcolor="#FF0000" style="color: #2c83b0;"><div
align="center" class="style16 style6
style21"><strong><strong>Patient
Name</strong></strong></div></td>
                        <td width="193" height="46" valign="middle"
bgcolor="#FF0000" style="color: #2c83b0;"><div
align="center" class="style16 style6

```

```

style21"><strong><strong>Attacker Name
</strong></strong></div></td>

<td width="128" height="46" valign="middle"
bgcolor="#FF0000" style="color: #2c83b0;"><div
align="center" class="style16 style6
style21"><strong><strong> Type
</strong></strong></div></td>

<td width="146" height="46" valign="middle"
bgcolor="#FF0000" style="color: #2c83b0;"><div
align="center" class="style16 style6
style21"><strong><strong>Date</strong></strong>
</div></td>

</tr>

<%@ include file="connect.jsp" %>

<%

```

```

String s1,s2,s3,s4,s5,s6,s7;
int i=0;
try
{
    String query="select * from attacker";
    Statement st=connection.createStatement();
    ResultSet rs=st.executeQuery(query);
    while ( rs.next() )
    {
        i=rs.getInt(1);
        s2=rs.getString(2);
        s3=rs.getString(3);//pn
        s4=rs.getString(4);
        s5=rs.getString(5);
        %>

        <tr>

        <td valign="baseline" height="0"><p
class="style22 style4 style5">&nbsp;</p>

```

```

<div align="center" class="style22 style4
style5">

    <%out.println(i);%>

    <p>&nbsp;</p>

</div></td>

```

```

<td valign="baseline" height="0"><p
class="style7 style4 style22">&nbsp;</p>

    <div align="center" class="style22 style4
style7 style9">

        <%out.println(s3);%>

        <p>&nbsp;</p>

    </div></td>

```

```

<td valign="baseline" height="0"><p
class="style7 style4 style22">&nbsp;</p>

    <div align="center" class="style22 style4
style7 style11">

        <%out.println(s2);%>

        <p>&nbsp;</p>

    </div></td>

```

```

<td valign="baseline" height="0"><p
class="style7 style4 style22">&nbsp;</p>

    <div align="center" class="style22 style4
style7 style13">

        <%out.println(s4);%>

        <p>&nbsp;</p>

    </div></td>

```

```

<td valign="baseline" height="0"><p class="style7
style4 style22">&nbsp;</p>

    <div align="center" class="style22 style4
style7 style15">

        <%out.println(s5);%>

        <p>&nbsp;</p>

    </div></td>

```

```

        </tr>
        <%
    }

```

```

connection.close();
}
catch(Exception e)
{
out.println(e.getMessage());
}
%>

```

```

        <tr>
            <td valign="baseline"
height="0">&nbsp;</td>
            <td valign="baseline"
height="0">&nbsp;</td>
            <td valign="baseline"
height="0">&nbsp;</td>
            <td valign="baseline"
height="0">&nbsp;</td>
            <td valign="baseline"
height="0">&nbsp;</td>
        </tr>
    </table>
    <p>&nbsp;</p>
    <p align="center"><a href="s_main.jsp"
class="style14">Back</a></p>
    <p>&nbsp;</p>
</div>
</div>
<div class="sidebar">
    <div class="gadget">

```



```

        <h2 class="star"><span>Server</span>
Menu</h2>
        <div class="clr"></div>
        <ul class="sb_menu">
            <li><a href="s_main.jsp"><span> Server
Main</span></a></li>
            <li><a href="cs_login.jsp"><span>Log
Out</span></a></li>
        </ul>
    </div>
</div>
<div class="clr"></div>
</div>
<div class="fbg"></div>
<div class="footer">
    <div class="footer_resize">
        <div style="clear:both;"></div>
    </div>
</div>
</div>
<div align=center></div>
</body>
</html>

```

```

<!DOCTYPE html PUBLIC "-//W3C//DTD
XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-
transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<title>Uploaded Patients</title>

```

```

<meta http-equiv="Content-Type"
content="text/html; charset=utf-8" />

<link href="css/style.css" rel="stylesheet"
type="text/css" />

<link rel="stylesheet" type="text/css"
href="css/coin-slider.css" />

<script type="text/javascript" src="js/cufon-
yui.js"></script>

<script type="text/javascript" src="js/cufon-
aller.js"></script>

<script type="text/javascript" src="js/jquery-
1.4.2.min.js"></script>

<script type="text/javascript"
src="js/script.js"></script>

<script type="text/javascript" src="js/coin-
slider.min.js"></script>

<script language="javascript"
type="text/javascript">

</script>

<style type="text/css">

<!--

.style1 {
font-size: 25px;
color: #33FF99;
}

.style2 {font-size: 25px}

-->

</style>

</head>

<body>

<div class="main">

  <div class="header">

    <div class="header_resize">

      <div class="logo">

```

```

        <h1><a href="index.html" class="style1
style2">Secure and Efficient Skyline Queries on
Encrypted Data <br />
</a></h1>
</div>
<div class="menu_nav">
    <ul>
        <li><a href="index.html"><span>Home
Page</span></a></li>
        <li><a
href="owner_login.jsp"><span>Owner</span></a>
</li>
        <li><a
href="user_login.jsp"><span>Client</span></a></l
i>
        <li class="active"><a
href="cs_login.jsp"><span>Cloud
Server</span></a></li>
        <li><a
href="au_login.jsp"><span></span></a></li>
    </ul>
</div>
<div class="clr"></div>
<div class="slider">
    <div id="coin-slider"> <a href="#"> </a> <a href="#"> </a> <a href="#"> </a> </div>
    <div class="clr"></div>
</div>
<div class="clr"></div>
</div>
</div>

```

```

<div class="content">
  <div class="content_resize">
    <div class="mainbar">
      <div class="article">
        <h2 align="center">View All Patient
Details</h2>
        <p align="center">&nbsp;</p>
        <p> <iframe src="patient_details2.jsp"
style="border:3px;" width="820"
height="300"></iframe> </p>
        <p>&nbsp;</p>
        <p align="center"><a href="s_main.jsp"
class="style14">Back</a></p>
        <p>&nbsp;</p>
      </div>
    </div>
  </div>
  <div class="sidebar">
    <div class="gadget">
      <h2 class="star"><span>Server</span>
Menu</h2>
      <div class="clr"></div>
      <ul class="sb_menu">
        <li><a href="s_main.jsp"><span>Server
Main</span></a></li>
        <li><a href="cs_login.jsp"><span>Log
Out</span></a></li>
      </ul>
    </div>
  </div>
  <div class="clr"></div>
</div>
<div class="fbg"></div>

```

```

<div class="footer">
  <div class="footer_resize">
    <div style="clear:both;"></div>
  </div>
</div>
</div>
<div align=center></div>
</body>
</html>

```

```

<!DOCTYPE html PUBLIC "-//W3C//DTD
XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-
transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<title> All Request and Resposes</title>
<meta http-equiv="Content-Type"
content="text/html; charset=utf-8" />
<link href="css/style.css" rel="stylesheet"
type="text/css" />
<link rel="stylesheet" type="text/css"
href="css/coin-slider.css" />
<script type="text/javascript" src="js/cufon-
yui.js"></script>
<script type="text/javascript" src="js/cufon-
aller.js"></script>
<script type="text/javascript" src="js/jquery-
1.4.2.min.js"></script>
<script type="text/javascript"
src="js/script.js"></script>
<script type="text/javascript" src="js/coin-
slider.min.js"></script>

```

```

<script language="javascript"
type="text/javascript">
</script>
<style type="text/css">
<!--
.style1 {
font-size: 25px;
color: #33FF99;
}
.style2 {font-size: 25px}
.style5 {font-size: 15px}
.style6 {color: #FFFFFF}
.style8 {font-weight: bold}
.style9 {font-size: 14px}
-->
</style>
</head>
<body>
<div class="main">
  <div class="header">
    <div class="header_resize">
      <div class="logo">
        <h1><a href="index.html" class="style1
style2">Secure and Efficient Skyline Queries on
Encrypted Data <br />
</a></h1>
      </div>
      <div class="menu_nav">
        <ul>
          <li><a href="index.html"><span>Home
Page</span></a></li>

```

```

        <li><a
href="owner_login.jsp"><span>Owner</span></a>
</li>

        <li><a
href="user_login.jsp"><span>Client</span></a></li>
i>

        <li class="active"><a
href="cs_login.jsp"><span>Cloud
Server</span></a></li>

        <li><a
href="au_login.jsp"><span></span></a></li>

    </ul>

</div>

<div class="clr"></div>

<div class="slider">

    <div id="coin-slider"> <a href="#"> </a> <a href="#"> </a> <a href="#"> </a> </div>

    <div class="clr"></div>

</div>

<div class="clr"></div>

</div>

<div class="content">

    <div class="content_resize">

        <div class="mainbar">

            <div class="article">

                <h2 align="center"> View All SK Requests
and Response </h2>

                <p align="center">&nbsp;</p>

```

```

<table width="694" border="1" align="center"
cellspacing="0" cellpadding="5">

  <tr>

    <td width="79" bgcolor="#FF0000"><div
align="center" class="style6 style1
style9"><strong>Id</strong></div></td>

    <td width="124" bgcolor="#FF0000"><div
align="center" class="style6 style1
style9"><strong>Patient
Name</strong></div></td>

    <td width="103" bgcolor="#FF0000"><div
align="center" class="style6 style1
style9"><strong>File Name</strong></div></td>

    <td width="116" bgcolor="#FF0000"><div
align="center" class="style6 style1
style9"><strong>Requested
Client</strong></div></td>

    <td width="113" bgcolor="#FF0000"><div
align="center" class="style6 style1
style9"><strong>Permission</strong></div></td>

    <td width="85" bgcolor="#FF0000"><div
align="center" class="style6 style1
style9"><strong>Date</strong></div></td>

  </tr>

```

```

<%@page
import="java.io.BufferedInputStream"%>

<%@page
import="java.security.DigestInputStream"%>

<%@page import="java.io.FileInputStream"%>

<%@page import="java.io.PrintStream"%>

<%@page import="java.io.FileOutputStream"%>

<%@page import="java.math.BigInteger"%>

<%@ page
import="java.security.Key,java.security.KeyPair,jav
a.security.KeyPairGenerator,javax.crypto.Cipher"%
>

```



```

<%@ include file="connect.jsp"%>

<%@pageimport="java.util.*,java.security.Key,java
.util.Random,javax.crypto.Cipher,javax.crypto.spec.
SecretKeySpec,org.bouncycastle.util.encoders.Base
64"%>

<%@page
import="java.security.MessageDigest"%>

<%@page import="java.sql.Statement"%>

<%@page import="java.sql.ResultSet"%>

<%@page
import="java.text.SimpleDateFormat"%>

<%@page import="java.util.Date"%>

<%

String s1 = "", s2 = "", s3 = "", s4 = "", s5 = "", s6 =
"", s7 = "", s8, s9 = "", s10, s11, s12,
s13,s14,s15,s16,s17;

int i = 0, j = 0, k = 0;

try {

String query = "select * from au_down_req ";
Statement st = connection.createStatement();
ResultSet rs = st.executeQuery(query);
while (rs.next()) {
i = rs.getInt(1);
s2 = rs.getString(2);
s3 = rs.getString(3);
s4 = rs.getString(4);
s5 = rs.getString(5);
s6 = rs.getString(6);

%> <tr>

```

```

<td height="29">
<div align="center" class="style8
style5"><%=i%></div>          </td>
<td height="29">
<div align="center" class="style8
style5"><%=s3%></div>          </td>
<td>
<div align="center" class="style8
style5"><%=s6%></div>          </td>
<td>
<div align="center" class="style8
style5"><%=s2%></div>          </td>
<td>
<div align="center" class="style8
style5"><%=s4%></div>          </td>
<td>
<div align="center" class="style8
style5"><%=s5%></div>          </td>

</tr>

<%
}

connection.close();
}

catch (Exception e)
{
out.println(e.getMessage());
}
%>
</table>

```

```

        <p align="center"><a href="s_main.jsp"
class="style14">Back</a></p>
    </div>
</div>
<div class="sidebar">
    <div class="gadget">
        <h2 class="star"><span>Server</span>
Menu</h2>
        <div class="clr"></div>
        <ul class="sb_menu">
            <li><a href="s_main.jsp"><span>Server
Main</span></a></li>
            <li><a href="cs_login.jsp"><span>Log
Out</span></a></li>
        </ul>
    </div>
</div>
<div class="clr"></div>
</div>
<div class="fbg"></div>
<div class="footer">
    <div class="footer_resize">
        <div style="clear:both;"></div>
    </div>
</div>
<div align=center></div>
</body>
</html>

```

```

<!DOCTYPE html PUBLIC "-//W3C//DTD
XHTML 1.0 Transitional//EN"

```

```

"http://www.w3.org/TR/xhtml1/DTD/xhtml1-
transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<title> All Transactions</title>
<meta http-equiv="Content-Type"
content="text/html; charset=utf-8" />
<link href="css/style.css" rel="stylesheet"
type="text/css" />
<link rel="stylesheet" type="text/css"
href="css/coin-slider.css" />
<script type="text/javascript" src="js/cufon-
yui.js"></script>
<script type="text/javascript" src="js/cufon-
aller.js"></script>
<script type="text/javascript" src="js/jquery-
1.4.2.min.js"></script>
<script type="text/javascript"
src="js/script.js"></script>
<script type="text/javascript" src="js/coin-
slider.min.js"></script>
<script language="javascript"
type="text/javascript">
</script>
<style type="text/css">
<!--
.style1 {
font-size: 25px;
color: #33FF99;
}
.style2 {font-size: 25px}
.style5 {font-size: 15px}
.style6 {color: #FFFFFF}
.style7 {font-size: 14px}

```

```

.style8 {font-weight: bold}
.style9 {font-weight: bold}
-->
</style>
</head>
<body>
<div class="main">
  <div class="header">
    <div class="header_resize">
      <div class="logo">
        <h1><a href="index.html" class="style1
style2">Secure and Efficient Skyline Queries on
Encrypted Data <br />
</a></h1>
      </div>
      <div class="menu_nav">
        <ul>
          <li><a href="index.html"><span>Home
Page</span></a></li>
          <li><a
href="owner_login.jsp"><span>Owner</span></a>
</li>
          <li><a
href="user_login.jsp"><span>Client</span></a></li>
          <li class="active"><a
href="cs_login.jsp"><span>Cloud
Server</span></a></li>
          <li><a
href="au_login.jsp"><span></span></a></li>
        </ul>
      </div>
      <div class="clr"></div>
      <div class="slider">

```

```

        <div id="coin-slider"> <a href="#"> </a> <a href="#"> </a> <a href="#"> </a> </div>

        <div class="clr"></div>

    </div>

    <div class="clr"></div>

</div>

<div class="content">
    <div class="content_resize">
        <div class="mainbar">
            <div class="article">
                <h2 align="center"> View All Transactions !!!
</h2>
<p align="center">&nbsp;</p>

<table width="694" border="1" align="center"
cellspacing="0" cellpadding="5">
    <tr>
        <td width="79" bgcolor="#FF0000"><div
align="center" class="style7 style6
style1"><strong>Id</strong></div></td>
        <td width="155" bgcolor="#FF0000"><div
align="center" class="style7 style6
style1"><strong>Client /
Owner</strong></div></td>
        <td width="175" bgcolor="#FF0000"><div
align="center" class="style1 style6 style7 style9">
            <p>Patient </p>
        </div></td>
    </tr>
</table>

```

```

<td width="141" bgcolor="#FF0000"><div
align="center" class="style7 style6
style1"><strong>Task</strong></div></td>

<td width="82" bgcolor="#FF0000"><div
align="center" class="style7 style6
style1"><strong>Date</strong></div></td>

</tr>

```

```

<%@page
import="java.io.BufferedInputStream"%>

<%@page
import="java.security.DigestInputStream"%>

<%@page import="java.io.FileInputStream"%>

<%@page import="java.io.PrintStream"%>

<%@page import="java.io.FileOutputStream"%>

<%@page import="java.math.BigInteger"%>

<%@ page
import="java.security.Key,java.security.KeyPair,jav
a.security.KeyPairGenerator,javax.crypto.Cipher"%
>

<%@ include file="connect.jsp"%>

<%@page
import="java.util.*,java.security.Key,java.util.Rand
om,javax.crypto.Cipher,javax.crypto.spec.SecretKe
ySpec,org.bouncycastle.util.encoders.Base64"%>

<%@page
import="java.security.MessageDigest"%>

<%@page import="java.sql.Statement"%>

<%@page import="java.sql.ResultSet"%>

<%@page
import="java.text.SimpleDateFormat"%>

<%@page import="java.util.Date"%>

<%

```

```
String s1 = "", s2 = "", s3 = "", s4 = "", s5 = "", s6 =
"", s7 = "", s8, s9 = "", s10, s11, s12,
s13,s14,s15,s16,s17;
int i = 0, j = 0, k = 0;
```

```
try {
```

```
String query = "select * from transaction ";
Statement st = connection.createStatement();
ResultSet rs = st.executeQuery(query);
while (rs.next()) {
i = rs.getInt(1);
s2 = rs.getString(2);
s3 = rs.getString(3);
s4 = rs.getString(4);
s5 = rs.getString(5);
//s6 = rs.getString(6);
```

```
%> <tr>
<td height="29">
<div align="center" class="style8
style5"><%=i%></div>          </td>
<td height="29">
<div align="center" class="style8
style5"><%=s2%></div>          </td>
<td>
<div align="center" class="style8
style5"><%=s3%></div>          </td>
<td>
<div align="center" class="style8
style5"><%=s4%></div>          </td>
<td>
```



```
<div align="center" class="style8
style5"><%=s5%></div>      </td>
```

```
/tr>
```

```
<%
}
```

```
connection.close();
}
```

```
catch (Exception e)
{
out.println(e.getMessage());
}
%>
</table>
```

```
    <p align="center"><a href="s_main.jsp"
class="style14">Back</a></p>
```

```
</div>
```

```
</div>
```

```
<div class="sidebar">
```

```
    <div class="gadget">
```

```
        <h2 class="star"><span>Server</span>
Menu</h2>
```

```
        <div class="clr"></div>
```

```
        <ul class="sb_menu">
```

```
            <li><a href="s_main.jsp"><span>Server
Main</span></a></li>
```

```
            <li><a href="cs_login.jsp"><span>Log
Out</span></a></li>
```

```
        </ul>
```

```

        </div>
    </div>
    <div class="clr"></div>
</div>
</div>
<div class="fbg"></div>
<div class="footer">
    <div class="footer_resize">
        <div style="clear:both;"></div>
    </div>
</div>
</div>
<div align=center></div>
</body>
</html>

```

```

<!DOCTYPE html PUBLIC "-//W3C//DTD
XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-
transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<title>Patient Disease Chart</title>
<meta http-equiv="Content-Type"
content="text/html; charset=utf-8" />
<link href="css/style.css" rel="stylesheet"
type="text/css" />
<link rel="stylesheet" type="text/css"
href="css/coin-slider.css" />
<script type="text/javascript" src="js/cufon-
yui.js"></script>
<script type="text/javascript" src="js/cufon-
aller.js"></script>

```

```

<script type="text/javascript" src="js/jquery-
1.4.2.min.js"></script>
<script type="text/javascript"
src="js/script.js"></script>
<script type="text/javascript" src="js/coin-
slider.min.js"></script>
<script language="javascript"
type="text/javascript">
</script>
<style type="text/css">
<!--
.style1 {
font-size: 25px;
color: #33FF99;
}
.style2 {font-size: 25px}
-->
</style>
</head>
<body>
<div class="main">
  <div class="header">
    <div class="header_resize">
      <div class="logo">
        <h1><a href="index.html" class="style1
style2">Secure and Efficient Skyline Queries on
Encrypted Data <br />
</a></h1>
      </div>
      <div class="menu_nav">
        <ul>
          <li><a href="index.html"><span>Home
Page</span></a></li>

```

```

        <li><a
href="owner_login.jsp"><span>Owner</span></a>
</li>

        <li><a
href="user_login.jsp"><span>Client</span></a></li>
i>

        <li class="active"><a
href="cs_login.jsp"><span>Cloud
Server</span></a></li>

        <li><a
href="au_login.jsp"><span></span></a></li>

    </ul>

</div>

<div class="clr"></div>

<div class="slider">

    <div id="coin-slider"> <a href="#"> </a> <a href="#"> </a> <a href="#"> </a> </div>

    <div class="clr"></div>

</div>

<div class="clr"></div>

</div>

<div class="content">

    <div class="content_resize">

        <div class="mainbar">

            <div class="article">

                <h2 align="center">View No.of Same
Disease in Chart</h2>

                <p align="center">&nbsp;</p>

            <p>

```

```

        <iframe align="middle" width="555"
height="400" src="graph1.jsp"></iframe></p>
        <p align="center"><a href="s_main.jsp"
class="style14">Back</a></p>
    </div>
</div>
<div class="sidebar">
    <div class="gadget">
        <h2 class="star"><span>Server</span>
Menu</h2>
        <div class="clr"></div>
        <ul class="sb_menu">
            <li><a href="s_main.jsp"><span>Server
Main</span></a></li>
            <li><a href="cs_login.jsp"><span>Log
Out</span></a></li>
        </ul>
    </div>
</div>
<div class="clr"></div>
</div>
<div class="fbg"></div>
<div class="footer">
    <div class="footer_resize">
        <div style="clear:both;"></div>
    </div>
</div>
</div>
<div align=center></div>
</body>
</html>

```

```

<!DOCTYPE html PUBLIC "-//W3C//DTD
XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-
transitional.dtd">

<html xmlns="http://www.w3.org/1999/xhtml">

<head>

<title>Patient Searched Rank Chart</title>

<meta http-equiv="Content-Type"
content="text/html; charset=utf-8" />

<link href="css/style.css" rel="stylesheet"
type="text/css" />

<link rel="stylesheet" type="text/css"
href="css/coin-slider.css" />

<script type="text/javascript" src="js/cufon-
yui.js"></script>

<script type="text/javascript" src="js/cufon-
aller.js"></script>

<script type="text/javascript" src="js/jquery-
1.4.2.min.js"></script>

<script type="text/javascript"
src="js/script.js"></script>

<script type="text/javascript" src="js/coin-
slider.min.js"></script>

<script language="javascript"
type="text/javascript">

</script>

<style type="text/css">

<!--

.style1 {
font-size: 25px;
color: #33FF99;
}

.style2 {font-size: 25px}

-->

</style>

```

```

</head>
<body>
<div class="main">
  <div class="header">
    <div class="header_resize">
      <div class="logo">
        <h1><a href="index.html" class="style1
style2">Secure and Efficient Skyline Queries on
Encrypted Data <br />
</a></h1>
      </div>
      <div class="menu_nav">
        <ul>
          <li><a href="index.html"><span>Home
Page</span></a></li>
          <li><a
href="owner_login.jsp"><span>Owner</span></a>
</li>
          <li><a
href="user_login.jsp"><span>Client</span></a></li>
          <li class="active"><a
href="cs_login.jsp"><span>Cloud
Server</span></a></li>
          <li><a
href="au_login.jsp"><span></span></a></li>
        </ul>
      </div>
      <div class="clr"></div>
      <div class="slider">
        <div id="coin-slider"> <a href="#"> </a> <a href="#"> </a> <a href="#"> </a> </div>
```

```
<div class="clr"></div>
```

```
</div>
```

```
<div class="clr"></div>
```

```
</div>
```

```
</div>
```

```
<div class="content">
```

```
<div class="content_resize">
```

```
<div class="mainbar">
```

```
<div class="article">
```

```
<h2 align="center">Patient Searched Rank in
Chart</h2>
```

```
<p align="center">&nbsp;</p>
```

```
<p> <iframe align="middle" width="555"
height="400" src="graph2.jsp"></iframe></p>
```

```
<p>&nbsp;</p>
```

```
<p align="center"><a href="s_main.jsp"
class="style14">Back</a></p>
```

```
<p>&nbsp;</p>
```

```
</div>
```

```
</div>
```

```
<div class="sidebar">
```

```
<div class="gadget">
```

```
<h2 class="star"><span>Server</span>
Menu</h2>
```

```
<div class="clr"></div>
```

```
<ul class="sb_menu">
```

```
<li><a href="s_main.jsp"><span>Server
Main</span></a></li>
```

```
<li><a href="cs_login.jsp"><span>Log
Out</span></a></li>
```



```

        </ul>
    </div>
</div>
<div class="clr"></div>
</div>
</div>
<div class="fbg"></div>
<div class="footer">
    <div class="footer_resize">
        <div style="clear:both;"></div>
    </div>
</div>
</div>
<div align=center></div>
</body>
</html>

```

```

<!DOCTYPE html PUBLIC "-//W3C//DTD
XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-
transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<title>Patient Download Rank Chart</title>
<meta http-equiv="Content-Type"
content="text/html; charset=utf-8" />
<link href="css/style.css" rel="stylesheet"
type="text/css" />
<link rel="stylesheet" type="text/css"
href="css/coin-slider.css" />
<script type="text/javascript" src="js/cufon-
yui.js"></script>

```

```

<script type="text/javascript" src="js/cufon-
aller.js"></script>

<script type="text/javascript" src="js/jquery-
1.4.2.min.js"></script>

<script type="text/javascript"
src="js/script.js"></script>

<script type="text/javascript" src="js/coin-
slider.min.js"></script>

<script language="javascript"
type="text/javascript">
</script>

<style type="text/css">
<!--
.style1 {
font-size: 25px;
color: #33FF99;
}
.style2 {font-size: 25px}
-->
</style>
</head>
<body>
<div class="main">
  <div class="header">
    <div class="header_resize">
      <div class="logo">
        <h1><a href="index.html" class="style1
style2">Secure and Efficient Skyline Queries on
Encrypted Data <br />
</a></h1>
      </div>
      <div class="menu_nav">
        <ul>

```

```

        <li><a href="index.html"><span>Home
Page</span></a></li>

        <li><a
href="owner_login.jsp"><span>Owner</span></a>
</li>

        <li><a
href="user_login.jsp"><span>Client</span></a></li>

        <li class="active"><a
href="cs_login.jsp"><span>Cloud
Server</span></a></li>

        <li><a
href="au_login.jsp"><span></span></a></li>
    </ul>
</div>

<div class="clr"></div>

<div class="slider">
    <div id="coin-slider"> <a href="#"> </a> <a href="#"> </a> <a href="#"> </a> </div>

    <div class="clr"></div>
</div>

<div class="clr"></div>
</div>

<div class="content">
    <div class="content_resize">
        <div class="mainbar">
            <div class="article">
                <h2 align="center">Patient File Downloaded
Rank in Chart</h2>

```

```

        <p align="center">&nbsp;</p>
    <p>
        <iframe align="middle" width="555"
height="400" src="graph3.jsp"></iframe></p>
        <p align="center"><a href="s_main.jsp"
class="style14">Back</a></p>
    <p>&nbsp;</p>
    </div>
</div>
<div class="sidebar">
    <div class="gadget">
        <h2 class="star"><span>Server</span>
Menu</h2>
        <div class="clr"></div>
        <ul class="sb_menu">
            <li><a href="s_main.jsp"><span>Server
Main</span></a></li>
            <li><a href="cs_login.jsp"><span>Log
Out</span></a></li>
        </ul>
    </div>
</div>
<div class="clr"></div>
</div>
<div class="fbg"></div>
<div class="footer">
    <div class="footer_resize">
        <div style="clear:both;"></div>
    </div>
</div>
</div>
<div align=center></div>

```

</body>

</html>

<!DOCTYPE html PUBLIC "-//W3C//DTD
XHTML 1.0 Transitional//EN"

"http://www.w3.org/TR/xhtml1/DTD/xhtml1-
transitional.dtd">

<html xmlns="http://www.w3.org/1999/xhtml">

<head>

<title>Patients Attacked Rank Chart</title>

<meta http-equiv="Content-Type"
content="text/html; charset=utf-8" />

<link href="css/style.css" rel="stylesheet"
type="text/css" />

<link rel="stylesheet" type="text/css"
href="css/coin-slider.css" />

<script type="text/javascript" src="js/cufon-
yui.js"></script>

<script type="text/javascript" src="js/cufon-
aller.js"></script>

<script type="text/javascript" src="js/jquery-
1.4.2.min.js"></script>

<script type="text/javascript"
src="js/script.js"></script>

<script type="text/javascript" src="js/coin-
slider.min.js"></script>

<script language="javascript"
type="text/javascript">

</script>

<style type="text/css">

<!--

.style1 {

font-size: 25px;

color: #33FF99;

}

```

.style2 {font-size: 25px}
-->
</style>
</head>
<body>
<div class="main">
  <div class="header">
    <div class="header_resize">
      <div class="logo">
        <h1><a href="index.html" class="style1
Secure and Efficient Skyline Queries on
Encrypted Data <br />
</a></h1>
      </div>
      <div class="menu_nav">
        <ul>
          <li><a href="index.html"><span>Home
Page</span></a></li>
          <li><a
href="owner_login.jsp"><span>Owner</span></a>
</li>
          <li><a
href="user_login.jsp"><span>Client</span></a>
</li>
          <li class="active"><a
href="cs_login.jsp"><span>Cloud
Server</span></a></li>
          <li><a
href="au_login.jsp"><span></span></a>
</li>
        </ul>
      </div>
      <div class="clr"></div>
      <div class="slider">

```

```

    <div id="coin-slider"> <a href="#"> </a> <a href="#"> </a> <a href="#"> </a> </div>

```

```

    <div class="clr"></div>

```

```

</div>

```

```

    <div class="clr"></div>

```

```

</div>

```

```

</div>

```

```

<div class="content">

```

```

    <div class="content_resize">

```

```

        <div class="mainbar">

```

```

            <div class="article">

```

```

                <h2 align="center">View No.Of Attackers on
Patient Accessing by Wrong Secret Key</h2>

```

```

                <p align="center">&nbsp;</p>

```

```

            <p>

```

```

                <iframe align="middle" width="555"
height="400" src="graph4.jsp"></iframe></p>

```

```

            <p align="center"><a
href="cs_all_attacker.jsp"
class="style14">Back</a></p>

```

```

        </div>

```

```

    </div>

```

```

<div class="sidebar">

```

```

    <div class="gadget">

```

```

        <h2 class="star"><span>Server</span>
Menu</h2>

```

```

    <div class="clr"></div>

```

```

    <ul class="sb_menu">

```

```

        <li><a href="s_main.jsp"><span>Server
Main</span></a></li>
        <li><a href="cs_login.jsp"><span>Log
Out</span></a></li>
    </ul>
</div>
</div>
<div class="clr"></div>
</div>
</div>
<div class="fbg"></div>
<div class="footer">
    <div class="footer_resize">
        <div style="clear:both;"></div>
    </div>
</div>
</div>
<div align=center></div>
</body>
</html>

```

```

<!DOCTYPE html PUBLIC "-//W3C//DTD
XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-
transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<title>Serevr LogIn</title>
<meta http-equiv="Content-Type"
content="text/html; charset=utf-8" />
<link href="css/style.css" rel="stylesheet"
type="text/css" />
<link rel="stylesheet" type="text/css"
href="css/coin-slider.css" />

```



```
<script type="text/javascript" src="js/cufon-yui.js"></script>
```

```
<script type="text/javascript" src="js/cufon-all.js"></script>
```

```
<script type="text/javascript" src="js/jquery-1.4.2.min.js"></script>
```

```
<script type="text/javascript" src="js/script.js"></script>
```

```
<script type="text/javascript" src="js/coin-slider.min.js"></script>
```

```
<script language="javascript" type="text/javascript">
```

```
function valid()
```

```
{
```

```
var na3=document.s.userid.value;
```

```
if(na3=="")
```

```
{
```

```
alert("Please Enter Name");
```

```
document.s.userid.focus();
```

```
return false;
```

```
}
```

```
else
```

```
{
```

```
}
```

```
var na4=document.s.pass.value;
```

```
if(na4=="")
```

```
{
```

```
alert("Please Enter Password");
```

```
document.s.pass.focus();
```

```
return false;
```

```

}

}
</script>
<style type="text/css">
<!--
.style1 {
font-size: 25px;
color: #33FF99;
}
.style2 {font-size: 25px}
.style3 {color: #FF0000}
-->
</style>
</head>
<body>
<div class="main">
  <div class="header">
    <div class="header_resize">
      <div class="logo">
        <h1><a href="index.html" class="style1
style2">Secure and Efficient Skyline Queries on
Encrypted Data <br />
</a></h1>
      </div>
      <div class="menu_nav">
        <ul>
          <li><a href="index.html"><span>Home
Page</span></a></li>
          <li><a
href="owner_login.jsp"><span>Owner</span></a>
</li>

```

```
        <li><a
href="user_login.jsp"><span>Client</span></a></li>
i>
```

```
        <li class="active"><a
href="cs_login.jsp"><span>Cloud
Server</span></a></li>
```

```
        <li><a
href="au_login.jsp"><span></span></a></li>
```

```
    </ul>
```

```
</div>
```

```
<div class="clr"></div>
```

```
<div class="slider">
```

```
    <div id="coin-slider"> <a href="#"> </a> <a href="#"> </a> <a href="#"> </a> </div>
```

```
    <div class="clr"></div>
```

```
</div>
```

```
<div class="clr"></div>
```

```
</div>
```

```
</div>
```

```
<div class="content">
```

```
    <div class="content_resize">
```

```
        <div class="mainbar">
```

```
            <div class="article">
```

```
                <h2 align="center"> Cloud Server Log
In</h2>
```

```
        <form name="s" action="s_authentication.jsp"
method="post" onSubmit="return valid()" ons
target="_top">
```

```
            <div align="center">
```

```

<table border="0" width="71%" height="165">
  <tr>
    <td height="25" colspan="2"><div
align="center"><em><strong></strong></em></div></td>
  </tr>
  <tr>
    <td width="48%" height="25"><span
class="style2 style3"><em><strong><font size="5"
face="Courier New, Courier, monospace">
Name</font> </strong></em></span></td>
    <td width="52%" height="25"><input
name="userid" type="text" size="25" /></td>
  </tr>
  <tr>
    <td width="48%" height="25"><span
class="style2 style3"><em><strong> <font size="5"
face="Courier New, Courier,
monospace">Pasword</font></strong></em></spa
n></td>
    <td width="52%" height="25"><input
name="pass" type="password" size="25" /></td>
  </tr>
  <tr>
    <td height="78" colspan="2"><p
align="center">
      <input type="submit" value="Login"
name="B1" />
      <input type="reset" value="Reset"
name="B2" />
    </p></td>
  </tr>
</table>

```

```

    </div>
</form>

    <p>&nbsp;</p>
</div>
</div>
<div class="sidebar">
    <div class="gadget">
        <h2 class="star"><span>Sidebar</span>
Menu</h2>
        <div class="clr"></div>
        <ul class="sb_menu">
            <li><a href="index.html"><span>Home
Page</span></a></li>
            <li><a
href="owner_login.jsp"><span>Owner</span></a>
</li>
            <li><a
href="user_login.jsp"><span>Client</span></a></li>
            <li><a href="cs_login.jsp"><span>Cloud
Server</span></a></li>
            <li><a
href="au_login.jsp"><span></span></a></li>
        </ul>
    </div>
</div>
<div class="clr"></div>
</div>
</div>
<div class="fbg"></div>
<div class="footer">
    <div class="footer_resize">
        <div style="clear:both;"></div>

```

```

        </div>
    </div>
</div>
<div align=center></div>
</body>
</html>

```

```

<%@page import="java.util.*"%>
<%@ include file="connect.jsp"%>
<%@page
import="java.util.*,java.security.Key,java.util.Rand
om,javax.crypto.Cipher,javax.crypto.spec.SecretKe
ySpec,org.bouncycastle.util.encoders.Base64"%>
<%@ page
import="java.sql.*,java.util.Random,java.io.PrintStr
eam,java.io.FileOutputStream,java.io.FileInputStrea
m,java.security.DigestInputStream,java.math.BigInt
eger,java.security.MessageDigest,java.io.BufferedIn
putStream"%>
<%@ page
import="java.security.Key,java.security.KeyPair,jav
a.security.KeyPairGenerator,javax.crypto.Cipher"%
>
<%@page
import="java.util.*,java.text.SimpleDateFormat,jav
a.util.Date,java.io.FileInputStream,java.io.FileOutp
utStream,java.io.PrintStream"%>

```

```

<%
    try {
        String fn =
request.getParameter("fname");
String pn = request.getParameter("pname");
        String strQuery = "select cont from
patient_details where pname='"+pn+ "' and
fname='"+fn+"' ";

```

```

        ResultSet rs =
connection.createStatement().executeQuery(strQuer
y);

        if (rs.next() == true)
{

String ct = rs.getString(1);//content
        String keys="q2e34rrfgfgfgg2a";
byte[] keyValue1 = keys.getBytes();

        Key key1 = new
SecretKeySpec(keyValue1, "AES");
Cipher c1 = Cipher.getInstance("AES");
c1.init(Cipher.DECRYPT_MODE, key1);


//String decryptedValue = new
String(Base64.decode(ct.getBytes()));

String file = new
String(Base64.decode(fn.getBytes()));


%>


</p>
<style type="text/css">
<!--
.style1 {
color: #FF0000;
font-weight: bold;
}
-->

</style>

```

```
<p align="center" class="style1 style11 style12
style1">Patient File Content(Encrypted)</p>
```

```
<td> <input type="hidden" id="fn"
value="<%=file %>" ></input></td>
```

```
<label>
```

```
<div align="center">
```

```
<textarea name="text" id="textarea"
cols="45" rows="17"><%=ct%></textarea>
```

```
<br/>
```

```
<br/>
```

```
<p>&nbsp;</p>
```

```
</div>
```

```
</label>
```

```
<p>
```

```
<%
```

```
}
```

```
else {
```

```
}
```

```
connection.close();
```

```
} //try close
```

```
catch (Exception e) {
```

```
out.println(e.getMessage());
```

```
}
```

```
%>
```


6.2 Implementation

The implementation phase is less creative than system design. It is primarily concerned with user training, and file conversion. The system may be requiring extensive user training. The initial parameters of the system should be modified as a result of a programming. A simple operating procedure is provided so that the user can understand the different functions clearly and quickly. The different reports can be obtained either on the inkjet or dot matrix printer, which is available at the disposal of the user. The proposed system is very easy to implement. In general implementation is used to mean the process of converting a new or revised system design into an operational one.

6.3 modules

1. Key Authorities
2. Storage Nodes
3. Sender
4. User

Modules Description:

Key Authorities:

They are key generation centers that generate public/secret parameters for CP-ABE. The key authorities consist of a central authority and multiple local authorities. We assume that there are secure and reliable communication channels between a central authority and each local authority during the initial key setup and generation phase. Each local authority manages different attributes and issues corresponding attribute keys to users. They grant differential access rights to individual users based on the users' attributes. The key authorities are assumed to be honest-but-curious. That is, they will honestly execute the assigned tasks in the system; however they would like to learn information of encrypted contents as much as possible.

Storage node:

This is an entity that stores data from senders and provide corresponding access to users. It may be mobile or static. Similar to the previous schemes, we also assume the storage node to be semitrusted that is honest-but-curious.

Sender:

This is an entity who owns confidential messages or data (e.g., a commander) and wishes to store them into the external data storage node for ease of sharing or for reliable delivery to users in the extreme networking environments. A sender is responsible for defining (attribute based) access policy and enforcing it on its own data by encrypting the data under the policy before storing it to the storage node.

User:

This is a mobile node who wants to access the data stored at the storage node (e.g., a soldier). If a user possesses a set of attributes satisfying the access policy of the encrypted data defined by the sender, and is not revoked in any of the attributes, then he will be able to decrypt the ciphertext and obtain the data.

7.SYSTEM TESTING

TESTING METHODOLOGIES

The following are the Testing Methodologies:

- **Unit Testing.**
- **Integration Testing.**
- **User Acceptance Testing.**
- **Output Testing.**
- **Validation Testing.**

7.1 Unit Testing

Unit testing focuses verification effort on the smallest unit of Software design that is the module. Unit testing exercises specific paths in a module's control structure to ensure complete coverage and maximum error detection. This test focuses on each module individually, ensuring that it functions properly as a unit. Hence, the naming is Unit Testing. During this testing, each module is tested individually and the module interfaces are verified for the consistency with design specification. All important processing path are tested for the expected results. All error handling paths are also tested.

7.2 Integration Testing

Integration testing addresses the issues associated with the dual problems of verification and program construction. After the software has been integrated a set of high order tests are conducted. The main objective in this testing process is to take unit tested modules and builds a program structure that has been dictated by design.

The following are the types of Integration Testing:

1. Top-Down Integration

This method is an incremental approach to the construction of program structure. Modules are integrated by moving downward through the control hierarchy, beginning with the main program module. The module subordinates to the main program module are incorporated into the structure in either a depth first or breadth first manner.

In this method, the software is tested from main module and individual stubs are replaced when the test proceeds downwards.

2. Bottom-up Integration

This method begins the construction and testing with the modules at the lowest level in the program structure. Since the modules are integrated from the bottom up, processing required for modules subordinate to a given level is always available and the need for stubs is eliminated. The bottom up integration strategy may be implemented with the following steps:

- The low-level modules are combined into clusters into clusters that perform a specific Software sub-function.
- A driver (i.e.) the control program for testing is written to coordinate test case input and output.
- The cluster is tested.
- Drivers are removed and clusters are combined moving upward in the program structure

The bottom up approaches tests each module individually and then each module is module is integrated with a main module and tested for functionality.

7.3 User Acceptance Testing

User Acceptance of a system is the key factor for the success of any system. The system under consideration is tested for user acceptance by constantly keeping in touch with the prospective system users at the time of developing and making changes wherever required. The system developed provides a friendly user interface that can easily be understood even by a person who is new to the system.

7.4 Output Testing

After performing the validation testing, the next step is output testing of the proposed system, since no system could be useful if it does not produce the required output in the specified format. Asking the users about the format required by them tests the outputs generated or displayed by the system under consideration. Hence the output format is considered in 2 ways – one is on screen and another in printed format.

7.5 Validation Checking

Validation checks are performed on the following fields.

Text Field:

The text field can contain only the number of characters lesser than or equal to its size. The text fields are alphanumeric in some tables and alphabetic in other tables. Incorrect entry always flashes and error message.

Numeric Field:

The numeric field can contain only numbers from 0 to 9. An entry of any character flashes an error message. The individual modules are checked for accuracy and what it has to perform. Each module is subjected to test run along with sample data. The individually tested modules are integrated into a single system. Testing involves executing the real data information is used in the program the existence of any program defect is inferred from the output. The testing should be planned so that all the requirements are individually tested. A successful test is one that gives

out the defects for the inappropriate data and produces an output revealing the errors in the system.

Preparation of Test Data

Taking various kinds of test data does the above testing. Preparation of test data plays a vital role in the system testing. After preparing the test data the system under study is tested using that test data. While testing the system by using test data errors are again uncovered and corrected by using above testing steps and corrections are also noted for future use.

Using Live Test Data

Live test data are those that are actually extracted from organization files. After a system is partially constructed, programmers or analysts often ask users to key in a set of data from their normal activities. Then, the systems person uses this data as a way to partially test the system. In other instances, programmers or analysts extract a set of live data from the files and have them entered themselves. It is difficult to obtain live data in sufficient amounts to conduct extensive testing. And, although it is realistic data that will show how the system will perform for the typical processing requirement, assuming that the live data entered are in fact typical, such data generally will not test all combinations or formats that can enter the system. This bias toward typical values then does not provide a true systems test and in fact ignores the cases most likely to cause system failure.

Using Artificial Test Data:

Artificial test data are created solely for test purposes, since they can be generated to test all combinations of formats and values. In other words, the artificial data, which can quickly be prepared by a data generating utility program in the information systems department, make possible the testing of all login and control paths through the program. The most effective test programs use artificial test data generated by persons other than those who wrote the programs. Often, an independent team of testers formulates a testing plan, using the systems specifications. The package “Virtual Private Network” has satisfied all the requirements specified as per software requirement specification and was accepted.

USER TRAINING

Whenever a new system is developed, user training is required to educate them about the working of the system so that it can be put to efficient use by those for whom the system has been primarily designed. For this purpose the normal working of the project was demonstrated to the prospective users. Its working is easily understandable and since the expected users are people who have good knowledge of computers, the use of this system is very easy.

7.6 Maintenance

This covers a wide range of activities including correcting code and design errors. To reduce the need for maintenance in the long run, we have more accurately defined the user's requirements during the process of system development. Depending on the requirements, this system has been developed to satisfy the needs to the largest possible extent. With development in technology, it may be possible to add many more features based on the requirements in future. The coding and designing is simple and easy to understand which will make maintenance easier.

Testing Strategies

A strategy for system testing integrates system test cases and design techniques into a well planned series of steps that results in the successful construction of software. The testing strategy must co-ordinate test planning, test case design, test execution, and the resultant data collection and evaluation. A strategy for software testing must accommodate low-level tests that are necessary to verify that a small source code segment has been correctly implemented as well as high level tests that validate major system functions against user requirements. Software testing is a critical element of software quality assurance and represents the ultimate review of specification design and coding. Testing represents an interesting anomaly for the software. Thus, a series of testing are performed for the proposed system before the system is ready for user acceptance testing.

System Testing:

Software once validated must be combined with other system elements (e.g. Hardware, people, database). System testing verifies that all the elements are proper and that overall system function performance is achieved. It also tests to find discrepancies between the system and its original objective, current specifications and system documentation

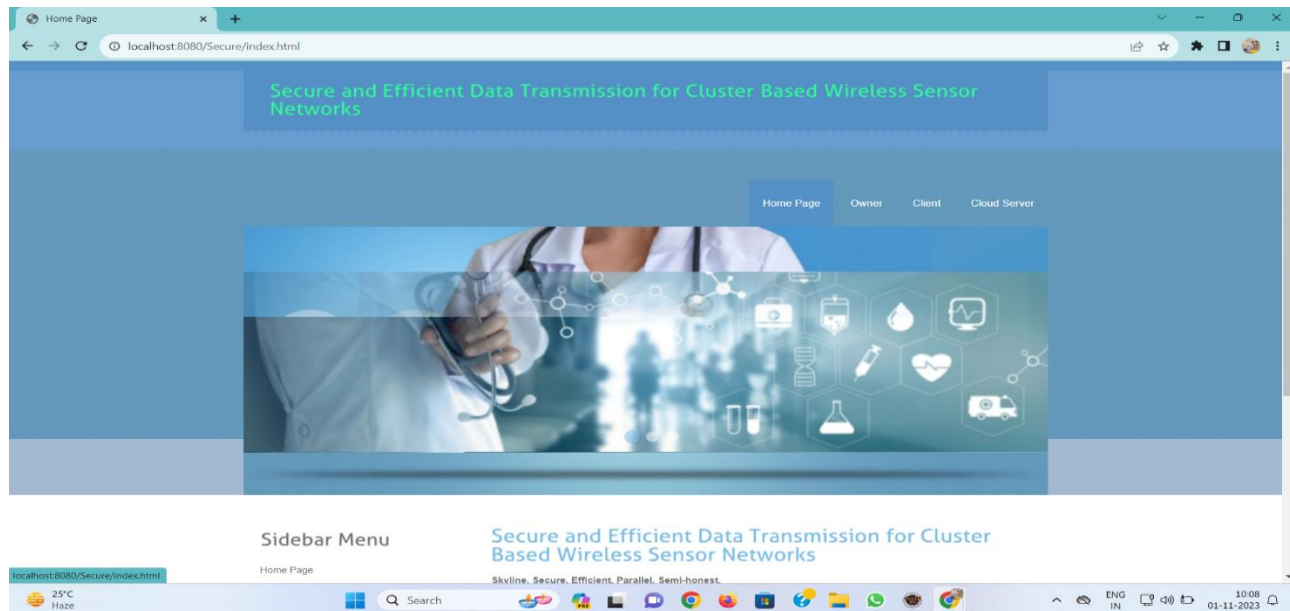
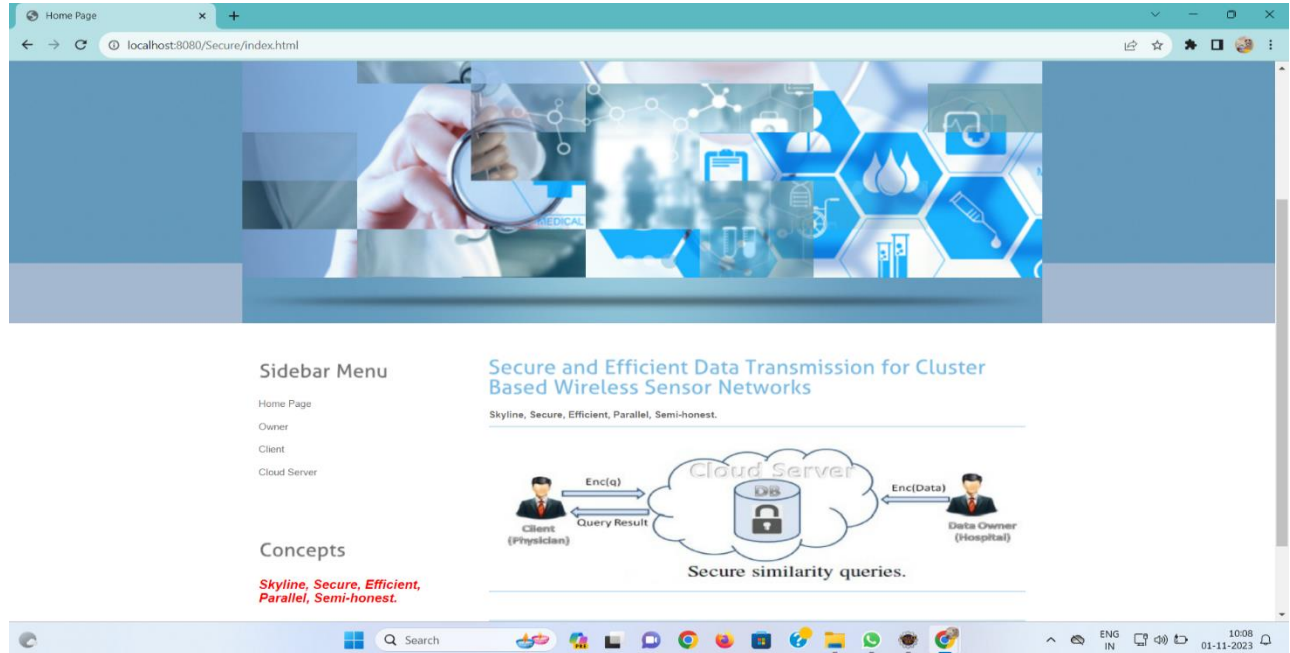
Unit Testing:

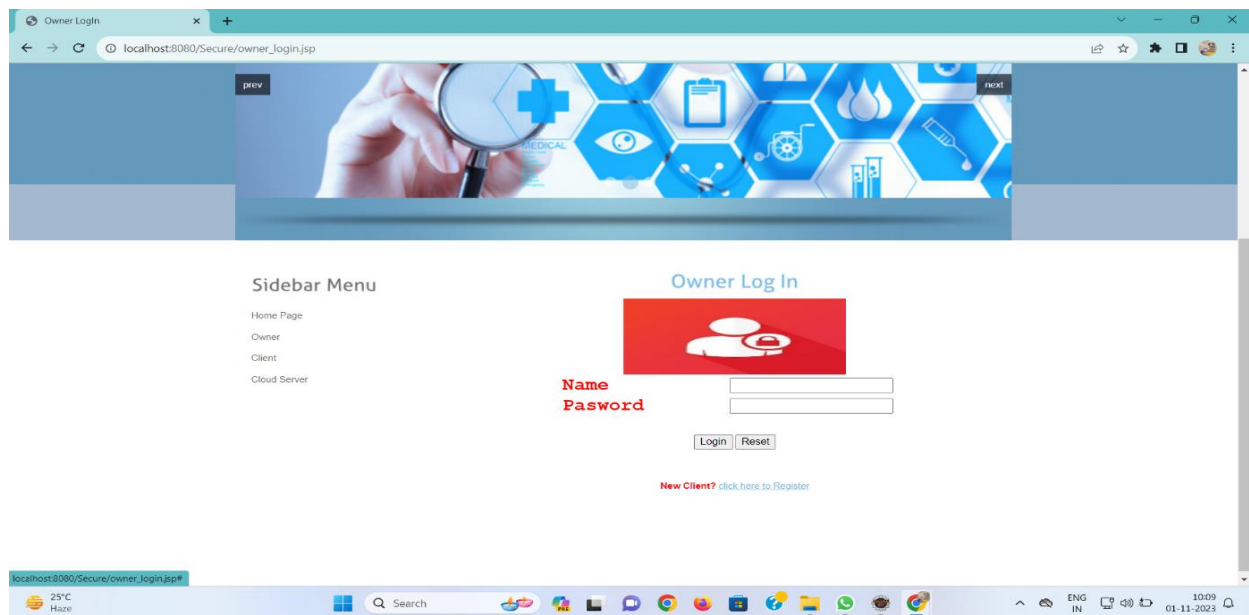
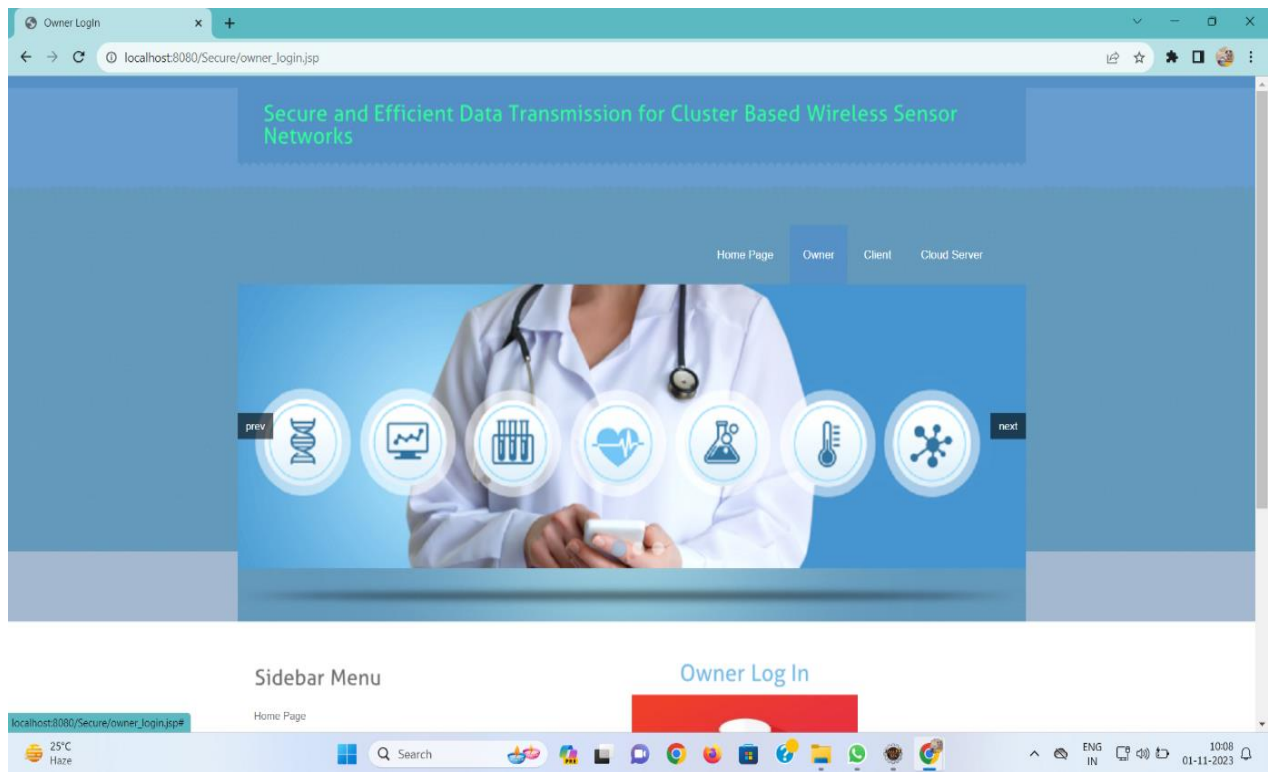
In unit testing different modules are tested against the specifications produced during the design for the modules. Unit testing is essential for verification of the code produced during the coding phase, and hence the goal is to test the internal logic of the modules. Using the detailed design description as a guide, important Conrail paths are tested to uncover errors within the boundary of the modules. This testing is carried out during the programming stage itself. In this type of testing step, each module was found to be working satisfactorily as regards to the expected output from the module. In Due Course, latest technology advancements will be taken into consideration. As part of technical build-up many components of the networking system will be generic in nature so that future projects can either use or interact with this. The future holds a lot to offer to the development and refinement of this project.

8. EXPERIMENTAL RESULTS

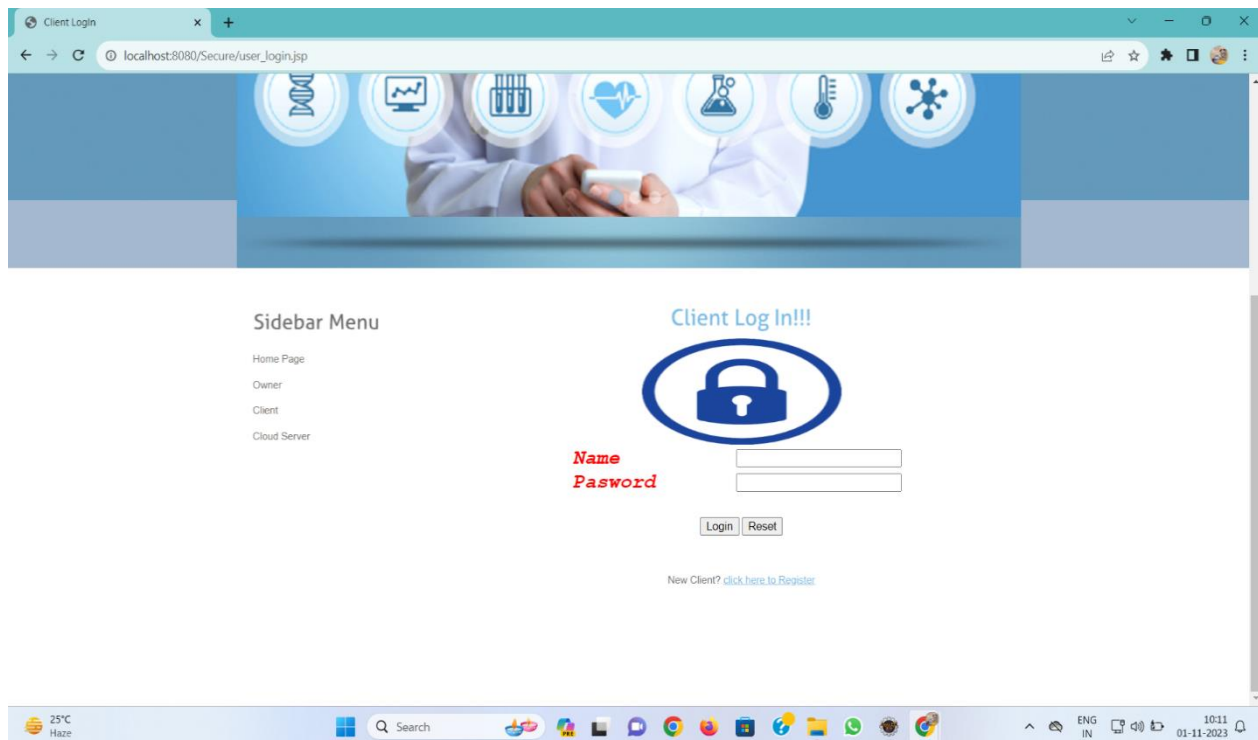
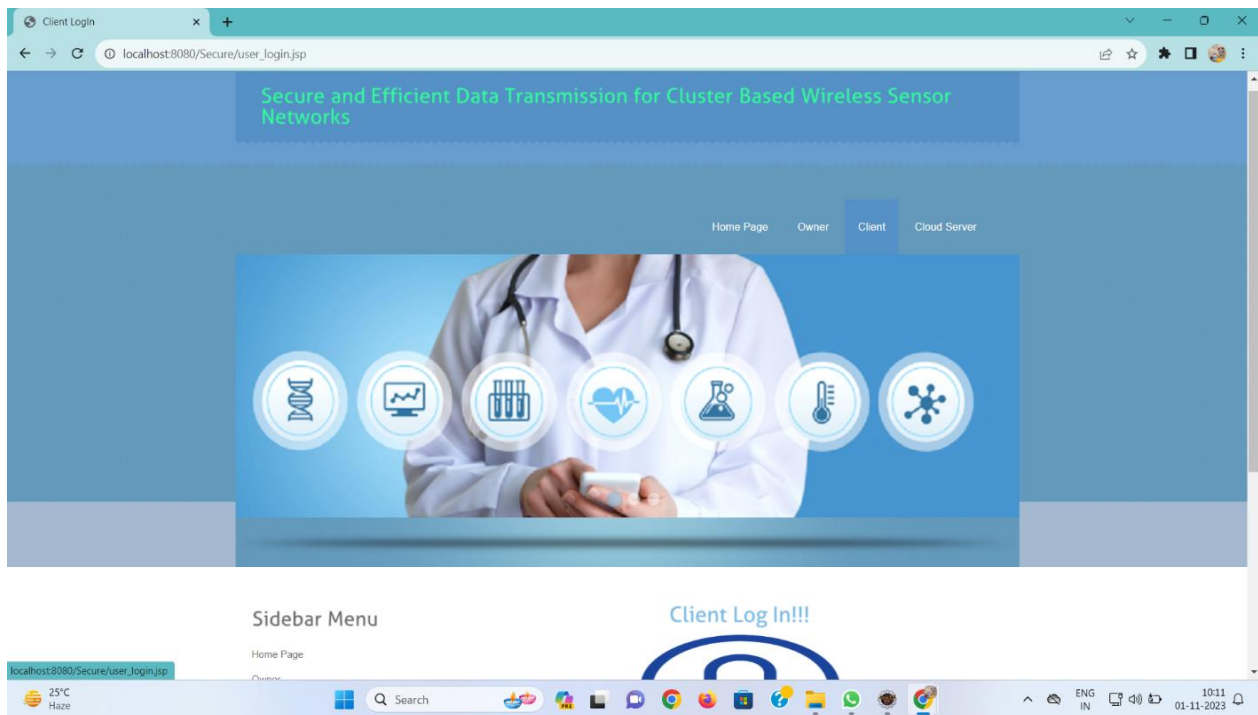
8.1 Screenshots

Welcome screen, login as administrator

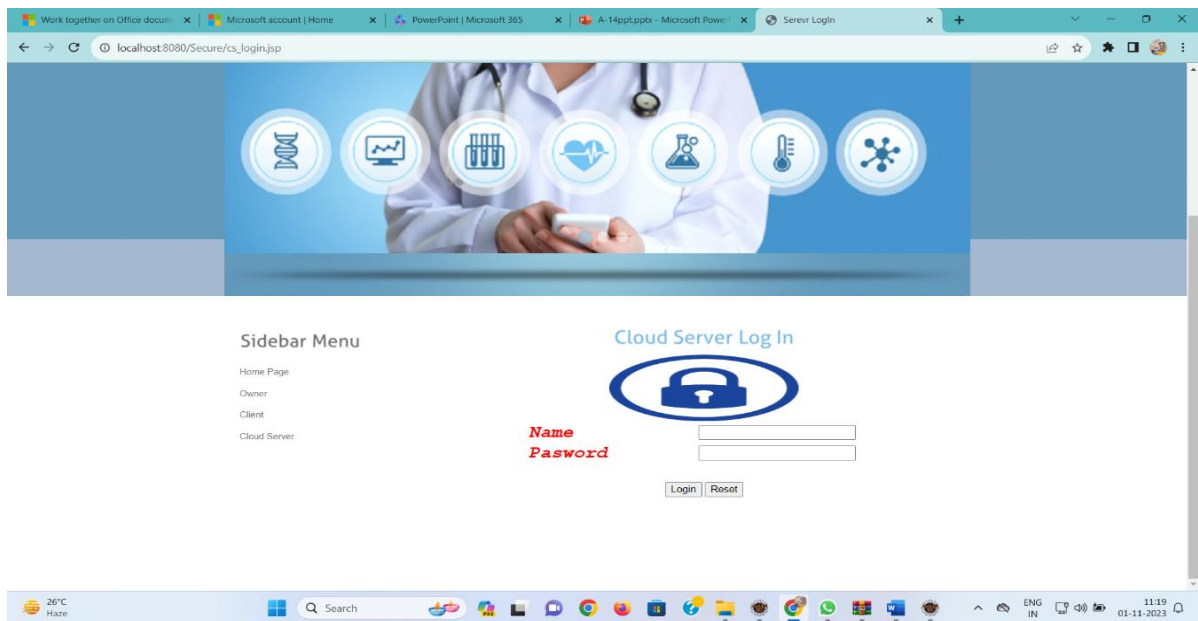




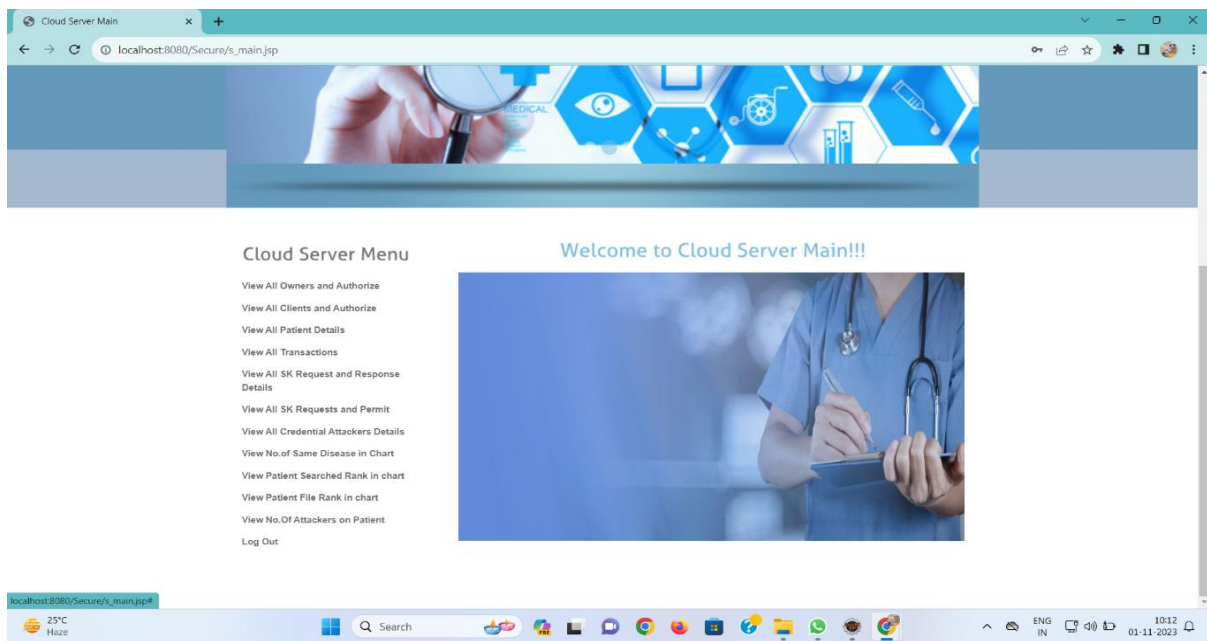
Owner login page



Client login page



Cloud Server login page



Cloud Server Main

View Owners

localhost:8080/Secure/au_all_owners.jsp

Menu

Main

Log Out

View All Owners and Authorise

ID	Owner Image	Owner Name	Department	Specilisation	Status
8		Kiran	Cardiology	Heart	Authorized
9		Manjunath	Cardiology	Heart	Authorized
10		Arun	Cardiology	Brain	Authorized
11		Ramesh	Cardiology	Heart	Authorized
12		pravalika	Cardiology	Heart	Authorized

25°C
Haze

Search

ENG
IN

10:12
01-11-2023

Owner and Authorisers details

View Clients

localhost:8080/Secure/au_all_users.jsp

Menu

Main

Log Out

View All Clients and Authorise!!!

ID	Client Image	Client Name	Department	Profession	Status
		Ramesh	Cardiology	Doctor	Authorized
		Imksmanju	Cardiology	Doctor	Authorized
		Uday	Neurology	Dentist	Authorized
		Krishnan	Cardiology	Surgeon	Authorized
		Suresh	Neurology	Doctor	Authorized

25°C
Haze

Search

ENG
IN

10:12
01-11-2023

Client and authorisers details

9. Future Enhancement

This project strongly focuses on ensuring the security of data transmission and processing in the context of cloud-based data storage and querying. As technology and research continue to advance, there is a significant future scope for this project. Here are some potential areas of future development and research

- **Advanced Encryption Techniques:** As encryption plays a pivotal role in ensuring data security, future work can explore advanced encryption techniques, including post-quantum cryptography, to enhance the security of data stored and transmitted in the cloud.
- **Performance Optimization:** Enhancing the efficiency and performance of the system can be a continuous effort. Researchers can work on optimizing the data partitioning and merging techniques to further reduce computational loads and improve system response times.
- **Scalability:** Scalability is essential in handling large-scale data storage and processing. Future research can focus on making the system more scalable to accommodate even larger datasets and higher query loads.
- **Privacy-Preserving Techniques:** Research can explore more advanced privacy-preserving techniques to protect sensitive information in various domains, such as healthcare, finance, and IoT applications.
- **Machine Learning Integration:** Integrating machine learning models into the system can help in more intelligent data processing and analytics. This can enable real-time decision-making based on encrypted data.
- **Multi-Party Computation (MPC):** Building on the secure multi-party computation (SMC) concept, future research can expand into multi-party settings and investigate applications in various domains, including secure collaborative data analysis.
- **Usability and User Experience:** Ensuring a seamless and user-friendly experience for end-users is crucial. Future work can focus on the usability and user experience design of the system.
- **Integration with Emerging Technologies:** As new technologies and paradigms emerge, such as edge computing and blockchain, integrating this project with these technologies could provide new dimensions of security and efficiency.

- **Cross-Domain Applications:** Expanding the project's applicability to various domains and industries beyond wireless sensor networks, such as finance, logistics, and smart cities, can be a valuable area of research.
- **Standardization and Compliance:** Future work can involve aligning the project with industry standards and compliance regulations to ensure that sensitive data handling is in accordance with legal and regulatory requirements.
- **Security Evaluation and Certification:** Conducting rigorous security evaluations and obtaining security certifications can enhance the credibility of the system and make it more appealing to organizations that require high levels of security.
- **Energy Efficiency:** In the context of wireless sensor networks, optimizing energy consumption for both data transmission and processing is essential. Future research can focus on developing energy-efficient solutions.
- **Integration with Cloud Services:** Integration with various cloud service providers and platforms can provide flexibility and options for users, making the system more versatile.

10. Conclusions

we proposed a fully secure skyline protocol on encrypted data using two non-colliding cloud servers under the semi-honest model. It ensures semantic security in that the cloud servers know nothing about the data including indirect data patterns, query, as well as the query result. In addition, the client and data owner do not need to participate in the computation. We also presented a secure dominance protocol which can be used by skyline queries as well as other queries. Furthermore, we demonstrated two optimizations, data partitioning and lazy merging, to further reduce the computation load. Finally, we presented our implementation of the protocol and demonstrated the feasibility and efficiency of the solution. As for future work, we plan to optimize the communication time complexity to further improve the performance of the protocol.

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